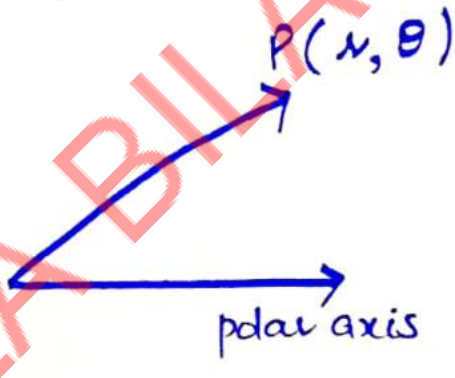
Lecture 23

Polar Coordinate System

Take a point O called origin/pole & draw a ray called polar axis.

Now take a point P with coordinates (r, θ) where "r" is distance of $\overline{OP}$ & θ is angle b/w $\overline{OP}$ & polar axis.



This is how we form a polar coordinate System.

Radial distance:-

The number r is called radial distance of P.

Polar angle:-

θ is called the polar angle of P.

P is written as $P(r, \theta)$ (e.g) $(6, 45^\circ), (4, 330^\circ)$ are plotted in polar coordinate System.

Remark. The points $(3, 225^\circ)$ & $(-3, 45^\circ)$ are same. (-ve sign shows that point is one extension of angle's terminal side rather than on terminal side itself. As $(\theta + 180)$ is extension of θ. So $(-r, \theta)$ can also be written as $(r, \theta + 180)$

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Relation b/w Polar & Rectangular coordinates:-

Polar coordinates $P(x, y)$

Rectangular Coordinates $P(r, \theta)$

$x = r\cos\theta$ & $y = r\sin\theta$

Conversion:-

From polar to Cartesian/Rectangular:-

$x = r\cos\theta$ & $y = r\sin\theta$

From Rectangular to Polar:-

$r^2 = x^2 + y^2$ & $\tan\theta = \frac{y}{x}$

Example:- Find rectangular coordinates of P whose polar coordinates are $(6, 135^\circ)$

$r = 6$, $\theta = 135^\circ$

As $x = r\cos\theta$

$= 6(\cos 135)$

$= 6\left(-\frac{\sqrt{2}}{2}\right)$

$= -3\sqrt{2}$

$y = r\sin\theta$

$y = 6\sin 135^\circ$

$= 6\left(\frac{\sqrt{2}}{2}\right)$

$y = 3\sqrt{2}$

So rectangular coordinates of P are

$(-3\sqrt{2}, 3\sqrt{2})$

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Example:- Find polar coordinates of P whose rectangular coordinates are $(-2, 2\sqrt{3})$.

here $x = -2$ & $y = 2\sqrt{3}$

$$r^2 = x^2 + y^2$$
$$= (-2)^2 + (2\sqrt{3})^2$$
$$= 4 + 4(3)$$
$$= 4 + 12$$
$$r^2 = 16$$
$$r = 4$$

$$\tan\theta = \frac{y}{x}$$
$$\tan\theta = \frac{2\sqrt{3}}{-2}$$
$$\tan\theta = (-\sqrt{3})$$
$$\theta = \tan^{-1}(-\sqrt{3})$$
$$\theta = \frac{2\pi}{3} \text{ or } 120^\circ$$

So the polar coordinates are (r, θ) or $(-r, \theta + 180)$

(i-e) $\left(4, \frac{2\pi}{3}\right), \left(-4, \frac{5\pi}{3}\right)$

Lines in Polar coordinates:-

If we have $x = a$ (i-e) point $(a, 0)$, the equation $x = r\cos\theta$ will be $a = r\cos\theta$.

If we have $y = b$ (ie) point $(0, b)$, the equation $y = r\sin\theta$ will be $b = r\sin\theta$.

If $\theta = \theta_0$ it means line passes through origin.

As general equation of line is $Ax + By + C = 0$.

Put values:- $A(r\cos\theta) + B(r\sin\theta) + C = 0$

$$r(A\cos\theta + B\sin\theta) + C = 0$$

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Circles in polar coordinates:-

If we have a circle of radius 'a' & its center is (r_0, θ_0), then by applying law of cosine

(general form:- $a^2 = b^2 + c^2 - 2bc\cos\theta$)

$$a^2 = r^2 - 2rr_0\cos(\theta - \theta_0) + r_0^2 \quad \text{---} \quad (1)$$

Special Cases:-

⇒ A circle of radius a & center at origin.

Let $r_0 = 0$ (put in eq 1)

$$a^2 = r^2 - 2r(0)\cos(\theta - \theta_0) + 0^2$$

$$a^2 = r^2$$

This circle consists of all points $P(r, \theta)$ for which $r = a$, regardless of θ.

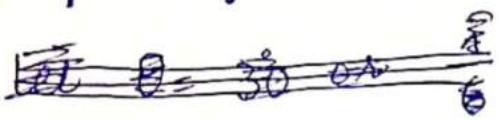
⇒ If circle of radius a & center on x-axis & passes through origin, then polar coordinates of center are either $(a, 0)$ or (a, π) depending on whether center is on right or left of origin.

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Lecture 24 Sketching

graph of the curve having equ $r = \sin\theta$



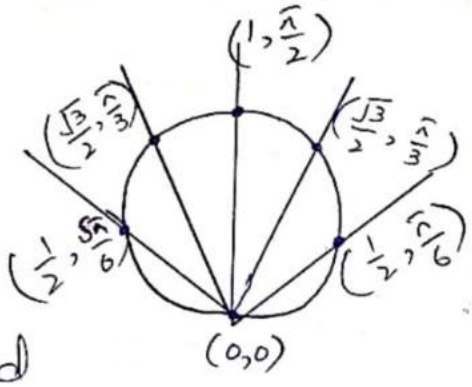
increase θ by 30° or $\frac{\pi}{6}$.

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
$r = \sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$

θ	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
$r = \sin\theta$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0

There are 13 pairs in Table but only 6 points are plotted bcz others are same as them.

(-ve value of r & θ is increased by 180° or π)



Equation of This circle:- As $r = \sin\theta$

(multiply b/s by r) $\quad r^2 = r\sin\theta$

($r^2 = x^2 + y^2$ & $y = r\sin\theta$) $\quad x^2 + y^2 = y$

$x^2 + y^2 - y = 0$

completing square $\quad x^2 + \left[y^2 - 2y\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2\right] - \left(\frac{1}{2}\right)^2 = 0$

$x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}$

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$$x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4} \quad \text{——— (1)}$$

As general eqn of circle is

$$(x - x_0)^2 + (y - y_0)^2 = r^2 \quad \left[(x_0, y_0) \text{ is center } \& \ r \text{ is radius}\right]$$

In eq (1) $x_0 = 0$ & $y_0 = -\frac{1}{2}$ | $r^2 = \frac{1}{4}$, $r = \frac{1}{2}$

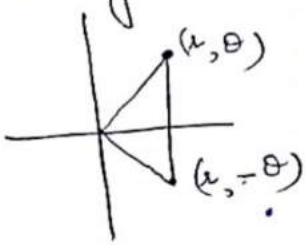
So center (x_0, y_0) is $\boxed{\left(0, -\frac{1}{2}\right)}$ & $\boxed{r = \frac{1}{2}}$

Curves in Polar Coordinates:-

1. Symmetry

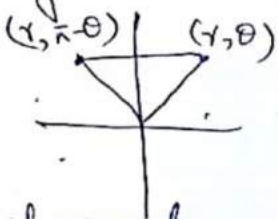
(i) about x-axis/ initial line:-

If equation of a line remains unchanged when (r, θ) is replaced by $(r, -\theta)$.



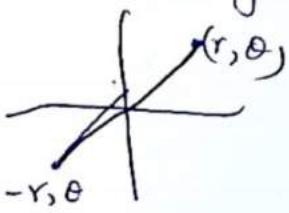
(ii) about y-axis:-

If equation of a line remains unchanged when (r, θ) is replaced by $(r, \pi - \theta)$



(iii) about Pole:-

If the equation of a line remains unchanged when (r, θ) is replaced by $(-r, \theta)$



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2. Position of the pole Relative to curve:-

Put $r=0$ & find value of θ for position.

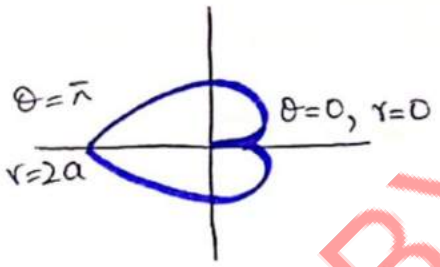
3. Table of values:-

Construct a sufficiently complete table of values.

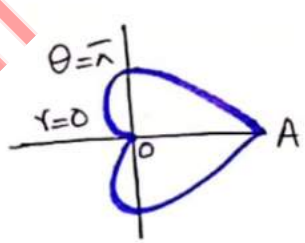
Position of Pole relative to curve:-

When $r=0$ & $\theta=0$ Then the curve passes through the pole.

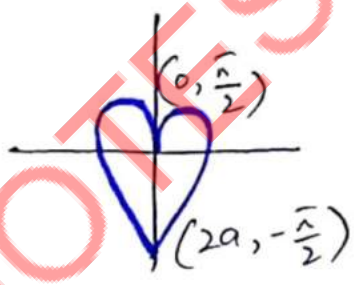
Cardioids:-



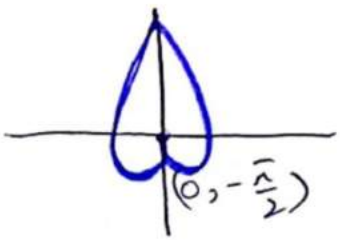
$r = a(1-\cos\theta)$



$r = a(1+\cos\theta)$



$r = a(1-\sin\theta)$



$r = a(1+\sin\theta)$

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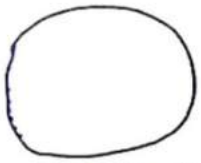
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Cardidoids & Limacons:-

$r = a + b\sin\theta$, $r = a - b\sin\theta$

$r = a + b\cos\theta$, $r = a - b\cos\theta$

These equation form polar curves called Limacons. If $a = b$ then above equations become same as for cardidoids. Their position depends upon whether $\sin\theta$ or $\cos\theta$ is +ve or -ve.



$\frac{a}{b} \geq 2$



$\frac{a}{b} < 1$

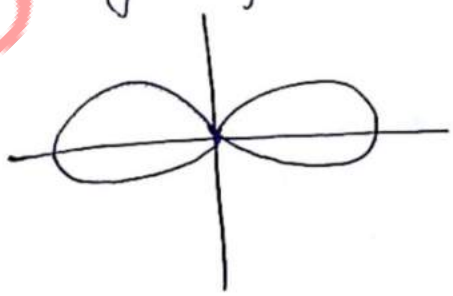
Lemicon with inner loop.

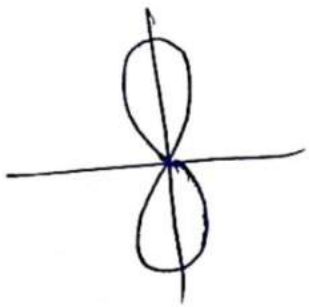
Leminscate:- if $a \neq 0$, then equation of form

$r^2 = a^2\cos 2\theta$, $r^2 = -a^2\cos 2\theta$

$r^2 = a^2\sin 2\theta$, $r^2 = -a^2\sin 2\theta$

represent propeller-shaped curves, called Leminscate. These are centered at origin. But position depends on sign of a^2.





(e.g) $r^2 = 4\cos 2\theta$ is eqn. of a leminscate.

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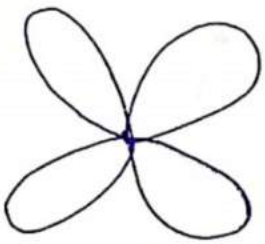
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Rose Curves:-

Equations of the form ~~$r = a \sin\theta$~~ $r = a \sin n\theta$ & $r = a \cos n\theta$ represent flower shaped curves called Roses.

The rose has n equally spaced petals or loops if n is odd & $2n$ equally spaced petals if n is even.



($n=2$, four petals)



($n=3$, Three petals)

orientation depends upon sign of a. & if $\sin\theta$ or $\cos\theta$ appears in equation.

Spiral:-

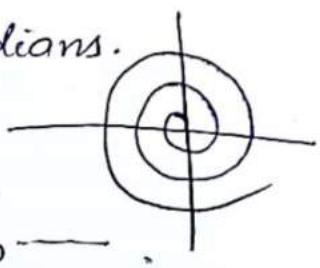
A curve that winds around the origin infinitely many times in such a way that r increases/decreases steadily as θ.

It has equations of form

$r = a\theta$ $(\theta \geq 0)$ or $r = a\theta$ $(\theta \leq 0)$

here a is +ve & θ is in radians.

its intersection with x-axis occurs at $\theta = 0, \pi, 2\pi, 3\pi, \ldots$ & intersection with y-axis occurs at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$



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Lecture 25

Double integral in Polar Coordinates

Double integral in Polar coordinates is given as

$$\iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

Example:- Find limits of integration for integrating $f(r,\theta)$ over Region R that lies inside the cardioid $r=1+\cos\theta$ & outside the circle $r=1$.

As

$$\iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

$r = 1+\cos\theta$

put $r=1$

$1 = 1+\cos\theta$

$1-1 = \cos\theta$

$0 = \cos\theta$

$\cos^{-1}(0) = \theta$

$\pm\frac{\pi}{2} = \theta$

So limits are

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_{1}^{1+\cos\theta} f(r,\theta)\,r\,dr\,d\theta.$$

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Example:- Evaluate $\iint_R \sin\theta \, dA$

where R is region in first quad. That is outside the circle $r=2$ & inside cardioid $r = 2(1+\cos\theta)$

As $r = 2(1+\cos\theta)$

put $r = 2$

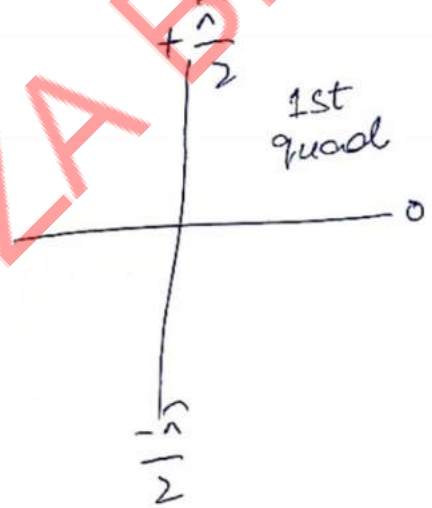
$2 = 2(1+\cos\theta)$

$1 = 1+\cos\theta$

$0 = \cos\theta$

$\cos^{-1}(0) = \theta$

$\pm \frac{\pi}{2} = \theta$



So $\int_0^{\pi/2} \int_2^{2(1+\cos\theta)} \sin\theta \, r \, dr \, d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\sin\theta \, \frac{r^2}{2} \right]_2^{2(1+\cos\theta)} d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\left(\sin\theta \, \frac{(2(1+\cos\theta))^2}{2} \right) - \left(\sin\theta \left(\frac{(2)^2}{2} \right) \right) \right] d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\left(\sin\theta \left(\frac{4(1+\cos\theta)^2}{2} \right) \right) - \left(\sin\theta \, \frac{4}{2} \right) \right] d\theta$

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$$= \int_0^{\frac{\pi}{2}} \left[2\sin\theta(1+\cos\theta)^2 - 2\sin\theta \right] d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left(-\sin\theta(1+\cos\theta)^2 + \sin\theta \right) d\theta$$

$$= 2\left[-\frac{(1+\cos\theta)^3}{3} + \cos\theta \right]_0^{\frac{\pi}{2}} + C$$

$$= 2\left[\left(-\frac{(1+\cos\frac{\pi}{2})^3}{3} + \cos\frac{\pi}{2} \right) - \left(-\frac{(1+\cos 0^\circ)^3}{3} + \cos 0^\circ \right) \right]$$

$$= 2\left[\left(-\frac{1}{3} + 0 \right) - \left(-\frac{8}{3} + 1 \right) \right]$$

$$= 2\left(-\frac{1}{3} + \frac{8}{3} - \frac{1}{1} \right)$$

$$= 2\left(\frac{-1+8-3}{3} \right) \Rightarrow 2\left(\frac{4}{3} \right)$$

$$= \frac{8}{3} \quad \text{Ans.}$$

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Example:- Use a double polar integral to find area enclosed by three-petaled rose $r = \sin 3\theta$. We calculate are in 1st quad & multiply by 3.

$$A = 3\iint_R dA$$

$$A = 3\iint_R r\,dr\,d\theta$$

$$= 3\int_0^{\frac{\pi}{3}}\int_0^{\sin 3\theta} r\,dr\,d\theta$$

$$= 3\int_0^{\frac{\pi}{3}}\left[\frac{r^2}{2}\right]_0^{\sin 3\theta} d\theta \Rightarrow \frac{3}{2}\int_0^{\frac{\pi}{3}} r^2\Big|_0^{\sin 3\theta} d\theta$$

$$= \frac{3}{2}\int_0^{\pi/3}\left((\sin^2 3\theta) - 0^2\right) d\theta$$

$$= \frac{3}{2}\int_0^{\frac{\pi}{3}} \frac{1-\cos 6\theta}{2}\, d\theta$$

$$= \frac{3}{2}\int_0^{\frac{\pi}{3}}\left(\frac{1}{2} - \frac{\cos 6\theta}{2}\right) d\theta \Rightarrow \frac{3}{4}\int_0^{\frac{\pi}{3}}(1-\cos 6\theta)\,d\theta$$

$$A = \frac{3}{4}\left(\theta - \frac{\sin 6\theta}{6}\right)\Big|_0^{\frac{\pi}{3}} \Rightarrow \frac{3}{4}\left(\frac{\pi}{3} - \frac{\sin 6\left(\frac{\pi}{3}\right)}{6}\right)$$

$$A = \frac{3}{4}\left(\frac{\pi}{3} - \frac{\sin 2\pi}{6}\right) \Rightarrow \frac{3}{4}\left(\frac{\pi}{3}\right) \Rightarrow \boxed{\frac{\pi}{4}}$$ Ans.

$r = \sin 3\theta$

put $r = 0$

$0 = \sin 3\theta$

$\sin^{-1}(0) = 3\theta$

$\pi = 3\theta$

$\frac{\pi}{3} = \theta$

$\because \sin^2\theta = \frac{1-\cos 2\theta}{2}$

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Find area enclosed by leminscate $r^2 = 4\cos 2\theta$. Total area is four times the first quad portion.

$r = 0$, $r^2 = 4\cos 2\theta$

$r = \sqrt{4\cos 2\theta}$

$0 = 4\cos 2\theta$

$0 = \cos 2\theta$

$\cos^{-1}(0) = 2\theta$

$\frac{\pi}{2} = \theta$

$\frac{\pi}{4} = \theta$, $\theta = 0$

$$A = 4\int_0^{\pi/4}\int_0^{\sqrt{4\cos 2\theta}} r\,dr\,d\theta \Rightarrow 4\int_0^{\frac{\pi}{4}} \frac{r^2}{2}\Big|_0^{\sqrt{4\cos 2\theta}} d\theta$$

$$= \frac{4}{2}\int_0^{\pi/4} \left(\sqrt{4\cos 2\theta}\right)^2 - 0^2 \; d\theta \Rightarrow 2\int_0^{\frac{\pi}{4}} 4\cos 2\theta\, d\theta$$

$$= 8\int_0^{\pi/4} \cos 2\theta\, d\theta \Rightarrow 8\left|\frac{\sin 2\theta}{2}\right|_0^{\frac{\pi}{4}}$$

$$= \frac{8}{2}\left(\sin 2\left(\frac{\pi}{4}\right) - \sin 2(0)\right)$$

$$= 4\left(\sin\frac{\pi}{2} - \sin 0\right)$$

$$= 4(1-0)$$

$$= 4(1)$$

$$\boxed{A = 4}$$ Ans.

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Changing Cartesian integral to polar.

Step 1: put $x = r\cos\theta$ & $y = r\sin\theta$ & replace $dx\,dy$ by $dr\,d\theta$.

Step 2: Supply polar limits of integration for R.

The integral becomes

$$\iint_R f(x,y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$$

Example:- Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)\,dy\,dx$ by changing into polar coordinates.

$0 \le y \le \sqrt{1-x^2}$ & $0 \le x \le 1$

$y = \sqrt{1-x^2}$

$y^2 = (\sqrt{1-x^2})^2$

$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

here $r^2 = 1$

so $r = 1$

$\theta = 0$ & $\theta = \frac{\pi}{2}$

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$$= \int_0^{\frac{\pi}{2}} \int_0^1 \left(r^2\cos^2\theta + r^2\sin^2\theta \right) r \, dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 r^2\left(\cos^2\theta + \sin^2\theta \right) r \, dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 r^2 (1) \; r \, dr \, d\theta$$

$$= \int_0^{\pi/2} \int_0^1 r^3 \, dr \, d\theta \Rightarrow \int_0^{\frac{\pi}{2}} \frac{r^4}{4} \Big|_0^1 d\theta$$

$$= \frac{1}{4} \int_0^{\pi/2} \left(1^4 - 0^4 \right) d\theta \Rightarrow \frac{1}{4} \int_0^{\pi/2} (1) \, d\theta$$

$$= \frac{1}{4} |\theta|_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{4}\left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{1}{4}\left(\frac{\pi}{2} \right) \Rightarrow \frac{\pi}{8} \quad \text{Ans.}$$

Example:- Evaluate $\iint_D \frac{dx\,dy}{x^2+y^2}$ by changing to polar coordinates where D is region of first quad b/w circles $x^2+y^2 = a^2$ & $x^2+y^2 = b^2$ $(0<a<b)$

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As $x^2+y^2=r^2$ if $a^2 \le r^2 \le b^2$

$a \le r \le b$

As quad is 1st So

$\theta = 0$ & $\theta = \frac{\pi}{2}$

$$= \int_0^{\frac{\pi}{2}} \int_a^b \frac{r\,dr\,d\theta}{r^2\cos^2\theta + r^2\sin^2\theta}$$

$$= \int_0^{\frac{\pi}{2}} \int_a^b \frac{r}{r^2}\,dr\,d\theta \Rightarrow \int_0^{\frac{\pi}{2}} \int_a^b \frac{1}{r}\,dr\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} |\ln r|_a^b\,d\theta \Rightarrow \int_0^{\pi/2} (\ln b - \ln a)\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \ln\left(\frac{b}{a}\right) d\theta \Rightarrow \ln\left(\frac{b}{a}\right) \int_0^{\pi/2} d\theta$$

$$= \ln\left(\frac{b}{a}\right) |\theta|_0^{\frac{\pi}{2}} \Rightarrow \ln\left(\frac{b}{a}\right)\left(\frac{\pi}{2} - 0\right)$$

$$= \frac{\pi}{2} \ln\left(\frac{b}{a}\right) \quad \text{Ans.}$$

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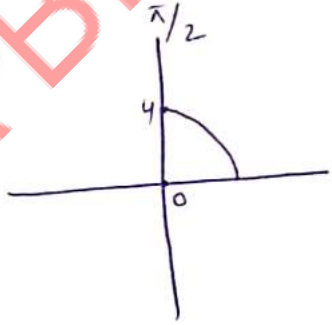
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Lecture No. 26

Examples:-

Example:- Evaluate $I = \int_0^4 \int_0^{\sqrt{4y-y^2}} (x^2+y^2)\, dx\, dy$ by changing into polar coordinates.

$0 \le x \le \sqrt{4y-y^2}$ & $0 \le y \le 4$



Take $x = \sqrt{4y-y^2}$

Take square on b/s

$x^2 = 4y - y^2$

$x^2 + y^2 = 4y$ (as general equ. of circle is $x^2+y^2 = r^2$)

So here

$r^2 = 4y$

$r^2 = 4r\sin\theta \quad (\because y = r\sin\theta)$

$r = 4\sin\theta$ & $r = 0$

$\theta = 0$ & $\theta = \frac{\pi}{2}$

Now $I = \int_0^{\frac{\pi}{2}} \int_0^{4\sin\theta} r^2 \, r\, dr\, d\theta \Rightarrow \int_0^{\pi/2} \int_0^{4\sin\theta} r^3\, dr\, d\theta$

$= \int_0^{\pi/2} \left| \frac{r^4}{4} \right|_0^{4\sin\theta} d\theta \Rightarrow \int_0^{\pi/2} \left(\frac{(4\sin\theta)^4}{4} - \frac{0^4}{4} \right) d\theta$

$= \int_0^{\pi/2} \frac{4^4 \sin^4\theta}{4}\, d\theta$

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$$= \int_0^{\pi/2} 4^3 \sin^4\theta \, d\theta$$

$$= \int_0^{\pi/2} 64 \sin^4\theta \, d\theta$$

Using Walli's formula

$$= \int_0^{\pi/2} 64 \frac{(4-1)(4-3)}{4(4-2)} d\theta \quad \because \frac{(m-1)(m-3)\cdots}{m(m-2)(m-4)\cdots}$$

$$= \int_0^{\pi/2} 64 \frac{(3)(1)}{4(2)} d\theta$$

$$= \int_0^{\pi/2} 64 \cdot \frac{3}{8} d\theta$$

$$= \int_0^{\pi/2} 24 \, d\theta \Rightarrow 24 \int_0^{\pi/2} d\theta$$

$$= 24 \, |\theta|_0^{\pi/2}$$

$$= 24 \left(\frac{\pi}{2} - 0\right)$$

$$= 24 \left(\frac{\pi}{2}\right)$$

$$= 12\pi \quad \text{Ans.}$$

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Example:- Evaluate $\iint_R e^{x^2+y^2}\, dy\, dx$ where R is semicircular region bounded by x-axis & the curve $y = \sqrt{1-x^2}$

$$y = \sqrt{1-x^2}$$

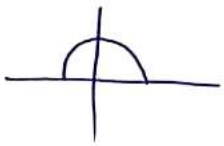
Take square on b/s

$$y^2 = 1 - x^2$$

$$y^2 + x^2 = 1 \quad \text{(comparing with } x^2+y^2=r^2\text{)}$$

here $r^2 = 1$, $r = 1$ & $r = 0$ (bcz semi circle

Now $\theta = 0$ & $\theta = \pi$ (bcz semi-circle on x-axis)



$$= \frac{1}{2}\int_0^{\pi}\int_0^{1} e^{r^2} 2r\, dr\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi} \left| e^{r^2} \right|_0^1 d\theta \Rightarrow \frac{1}{2}\int_0^{\pi} (e^1 - e^0)\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi} (e-1)\, d\theta \Rightarrow \frac{1}{2}(e-1)\int_0^{\pi} d\theta$$

$$= \frac{1}{2}(e-1)\,(\theta)\Big|_0^{\pi} \Rightarrow \frac{1}{2}(e-1)(\pi - 0)$$

$$= \frac{\pi}{2}(e-1) \quad \text{Ans.}$$

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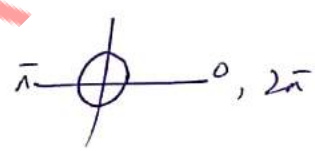
Example:- Let R_a be region bounded by the circle $x^2+y^2=a^2$. Define

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy = \lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}\,dx\,dy$$

$x^2+y^2=a^2$ here $r^2=a^2$

$r=a$ & $r=0$

$\theta=0$ & $\theta=2\pi$ (bcz region is bounded by circle)



$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy = \lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}\,dx\,dy$$

$$= \lim_{a\to\infty} -\frac{1}{2}\int_0^{2\pi}\int_0^{a} e^{-r^2}(-2)r\,dr\,d\theta$$

$$= \frac{-1}{2}\lim_{a\to\infty}\int_0^{2\pi}\left|e^{-r^2}\right|_0^a d\theta$$

$$= \frac{-1}{2}\lim_{a\to\infty}\int_0^{2\pi}\left(e^{-a^2}-e^{-0}\right)d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}(e^{-a^2}-1)\,d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}(e^{-a^2}-1)\int_0^{2\pi} d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}(e^{-a^2}-1)|\theta|_0^{2\pi}$$

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$$= -\frac{1}{2} \lim_{a \to \infty} \left(e^{-a^2} - 1\right)\left(2\bar{\pi} - 0\right)$$

$$= \lim_{a \to \infty} \left(1 - e^{-a^2}\right)\left(\frac{2\bar{\pi}}{2}\right)$$

$$= \lim_{a \to \infty} \left(1 - \frac{1}{e^{a^2}}\right)\bar{\pi}$$

$$= \left(1 - \frac{1}{e^{\infty}}\right)\bar{\pi} \Rightarrow \left(1 - \frac{1}{\infty}\right)\bar{\pi}$$

$$= (1 - 0)\,\bar{\pi} \Rightarrow \bar{\pi} \quad \underline{\text{Ans.}}$$

Example: Prove That $\int_0^{\infty} e^{-t^2}\,dt = \frac{1}{2}\sqrt{\bar{\pi}}$

Can't solve directly bcz derivative of $-t^2$ is not given. So

let $I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2-y^2}\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2}\cdot e^{-y^2}\,dx\,dy$$

$$= \left[\int_{-\infty}^{\infty} e^{-x^2}\,dx\right]\left[\int_{-\infty}^{\infty} e^{-y^2}\,dy\right]$$

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use a dummy variable t.

$$= \left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]\left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]$$

$$= \left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]^2$$

introduce a vawable a & apply limit

$$= \lim_{a\to\infty}\left(\int_{-a}^{a} e^{-t^2}\,dt\right)^2$$

$$= \lim_{a\to\infty} 4\left(\int_{0}^{a} e^{-t^2}\,dt\right)^2 \qquad \left(\int_0^a \text{ shows one quadrant}\right)$$

Here

$$4\lim_{a\to\infty}\left(\int_0^a e^{-t^2}\,dt\right)^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$$

$$4\lim_{a\to\infty}\left(\int_0^a e^{-t^2}\,dt\right)^2 = \pi \qquad \text{(reference to previous example).}$$

$$\left(\int_0^{\infty} e^{-t^2}\,dt\right)^2 = \frac{\pi}{4}$$

$$\int_0^{\infty} e^{-t^2}\,dt = \frac{\sqrt{\pi}}{2} \text{ or } \frac{1}{2}\sqrt{\pi}$$

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Theorem:- Let G be a rectangular box defined by inequalities

$$a \le x \le b \;,\; c \le y \le d \;,\; k \le z \le l$$

If G is continuous then

$$\iiint_G f(x,y,z)\,dV = \int_a^b \int_c^d \int_k^l f(x,y,z)\,dz\,dy\,dx$$

order of integration doesn't matter.

Example:- Evaluate Tripple integral

$\iiint_G 12xy^2z^3\,dV$ over rectangular box G defined by inequalities

$$-1 \le x \le 2 \;,\; 0 \le y \le 3 \;,\; 0 \le z \le 2.$$

$$= \int_{-1}^{2}\int_{0}^{3}\int_{0}^{2} 12xy^2z^3\,dz\,dy\,dx$$

$$= \int_{-1}^{2}\int_{0}^{3} 12xy^2 \left|\frac{z^4}{4}\right|_0^2 dy\,dx$$

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$$= \int_{-1}^{2} \int_{0}^{3} \frac{12xy^2}{4} (2^4 - 0^4)\, dy\, dx$$

$$= \int_{-1}^{2} \int_{0}^{3} 3xy^2 (16)\, dy\, dx$$

$$= \int_{-1}^{2} \int_{0}^{3} 48xy^2\, dy\, dx$$

$$= \int_{-1}^{2} 48x \left| \frac{y^3}{3} \right|_0^3 dx$$

$$= \int_{-1}^{2} \frac{48x}{3} (3^3 - 0^3)\, dx$$

$$= \int_{-1}^{2} 16x (27)\, dx$$

$$= \int_{-1}^{2} 432x\, dx$$

$$= 432 \left| \frac{x^2}{2} \right|_{-1}^{2} dx$$

$$= \frac{432}{2} (2^2 - (-1)^2) \Rightarrow 216(4-1)$$

$$\Rightarrow 216(3) \Rightarrow 648 \text{ Ans.}$$

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Example:- $\iiint_R (x-2y+z)\,dx\,dy\,dz$ when

$0 \le x \le 1$, $0 \le y \le x^2$, $0 \le z \le x+y$

$$\int_0^1 \int_0^{x^2} \int_0^{x+y} (x-2y+z)\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left| \frac{(x-2y+z)^2}{2} \right|_0^{x+y} dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left[\frac{(x-2y+x+y)^2}{2} - \frac{(x-2y)^2}{2} \right] dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{(2x-y)^2 - (x-2y)^2}{2} \right) dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{4x^2+y^2-4xy - x^2-4y^2+4xy}{2} \right) dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{x^2} (3x^2-3y^2)\,dy\,dx$$

$$= \frac{3}{2} \int_0^1 \int_0^{x^2} (x^2-y^2)\,dy\,dx$$

$$= \frac{3}{2} \int_0^1 \left(x^2y - \frac{y^3}{3} \right)\Big|_0^{x^2} dx$$

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$$= \frac{3}{2}\int_0^1 \left(\left[x^2(x^2) - \frac{(x^2)^3}{3} \right] - \left[x^2(0) - \frac{0^3}{3} \right] \right) dx$$

$$= \frac{3}{2}\int_0^1 \left(x^4 - \frac{x^6}{3} \right) dx$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{3.7} \right)\Big|_0^1$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{21} \right)\Big|_0^1$$

$$= \frac{3}{2}\left[\left(\frac{(1)^5}{5} - \frac{(1)^7}{21} \right) - \left(\frac{(0)^5}{5} - \frac{(0)^7}{21} \right) \right]$$

$$= \frac{3}{2}\left[\frac{1}{5} - \frac{1}{21} \right]$$

$$= \frac{3}{2}\left[\frac{21-5}{105} \right]$$

$$= \frac{3}{2}\left(\frac{16}{105} \right)$$

$$= \frac{8}{35} \quad \text{Ans.}$$

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Example:- Evaluate $\iiint xyz \, dx\,dy\,dz$

where $S = \{(x,y,z) : x^2+y^2+z^2 \le 1,\ x \ge 0, y \ge 0, z \ge 0\}$

$x^2+y^2+z^2=1$

$z^2 = 1-x^2-y^2$

$z = \sqrt{1-x^2-y^2}$

ignore z

$x^2+y^2=1$

$y^2 = 1-x^2$

$y = \sqrt{1-x^2}$

ignore y.

$x^2 = 1$

$x = 1$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz\,dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left| \frac{z^2}{2} \right|_0^{\sqrt{1-x^2-y^2}} dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \frac{\left(\sqrt{1-x^2-y^2}\right)^2}{2} dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} xy(1-x^2-y^2) \, dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} x\left(y - x^2y - y^3\right) dy\,dx$$

$$= \frac{1}{2} \int_0^1 \left[x \left| \frac{y^2}{2} - \frac{x^2y^2}{2} - \frac{y^4}{4} \right|_0^{\sqrt{1-x^2}} \right] dx$$

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$$= \frac{1}{2}\int_0^1 x\left(\frac{\left(\sqrt{1-x^2}\right)^2}{2} - \frac{x^2\left(\sqrt{1-x^2}\right)^2}{2} - \frac{\left(\sqrt{1-x^2}\right)^4}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-x^2}{2} - \frac{x^2(1-x^2)}{2} - \frac{(1-x^2)^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-x^2}{2} - \frac{x^2-x^4}{2} - \frac{1+x^4-2x^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{2-2x^2-2x^2+2x^4-1-x^4+2x^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-2x^2+x^4}{4}\right) dx$$

$$= \frac{1}{8}\int_0^1 \frac{x-2x^3+x^5}{} dx$$

$$= \frac{1}{8}\int_0^1 x^5 - 2x^3 + x \; dx$$

$$= \frac{1}{8}\left[\frac{x^6}{6} - \frac{2x^4}{4} + \frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{8}\left[\frac{(1)^6}{6} - \frac{2(1)^4}{4} + \frac{(1)^2}{2}\right] - 0$$

$$= \frac{1}{8}\left(\frac{1}{6} - \frac{2}{4} + \frac{1}{2}\right) \Rightarrow \frac{1}{8}\left(\frac{1-3+3}{6}\right) \Rightarrow \frac{1}{12}$$ Ans.

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Lecture 27

Vector Valued Functions

Those functions for which the domain consists of real numbers & the range consists of vectors in 2d-space & or 3d-space are called Vector valued functions.

Such functions can be expressed in

2d-space like $r(t) = (x(t), y(t))$

$= x(t)i + y(t)j$

Such functions can be expressed in

3d-space like $r(t) = (x(t), y(t), z(t))$

$= x(t)i + y(t)j + z(t)k$

here $x(t)$, $y(t)$ & $z(t)$ are called real valued functions of the real variable t. These functions are called component functions or components of r.

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Representation:-

Vector valued functions are represented by bold letters $\mathbf{f}(t)$, $\mathbf{g}(t)$ etc

Real valued functions are represented by italic lightface type. $f(t)$, $g(t)$ etc.

Parametric Equations in vector Form:-

Vector valued functions can be used to express parametric equations in 2-space/ 3-space in a compact Form.

For example consider parametric equations

$x = x(t)$ & $y = y(t)$.

Two vectors are equivalent iff their corresponding components are equal, so this pair of equations can be replaced by single vector equation.

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$x = x(t)$, $y = y(t)$

$x\underline{i} + y\underline{j} = x(t)\underline{i} + y(t)\underline{j}$

Similarly for Three dimensional space These equations

$x = x(t)$, $y = y(t)$, $z = z(t)$

can be replaced by single vector equation

$$x\underline{i} + y\underline{j} + z\underline{k} = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$$

In 2-d

$r = x\underline{i} + y\underline{j}$ & $r(t) = x(t)\underline{i} + y(t)\underline{j}$ are same.

In 3-d

$r = x\underline{i} + y\underline{j} + z\underline{k}$ & $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$

are same.

So we can write $r = r(t)$.

Example:- Express given parametric equ. in single vector equ.

(a) $x = t^2$ & $y = 3t$

$$r = x\underline{i} + y\underline{j} = t^2\underline{i} + 3t\underline{j}$$

(b) $x = \cos t$, $y = \sin t$, $z = t$

$$r = x\underline{i} + y\underline{j} + z\underline{k} = \cos t\,\underline{i} + \sin t\,\underline{j} + t\,\underline{k}$$

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Example:- Find parametric eqn. that correspond to the vector equation.

$$x\underline{i} + y\underline{j} + z\underline{k} = (t^3+1)\underline{i} + 3\underline{j} + e^t\underline{k}$$

Comparing

$$x = t^3+1 \;,\quad y = 3 \;,\quad z = e^t$$

Example:- Sketch graph of vector-valued function $r(t) = \cos t\,\underline{i} + \sin t\,\underline{j} \quad 0 \le t \le 2\pi$

As $r(t) = x\underline{i} + y\underline{j}$

$\cos t\,\underline{i} + \sin t\,\underline{j} = x\underline{i} + y\underline{j}$

Comparing

$$x = \cos t \;\&\; y = \sin t \quad (0 \le t \le 2\pi)$$

Take square on b/s

$$x^2 = \cos^2 t \;\&\; y^2 = \sin^2 t$$

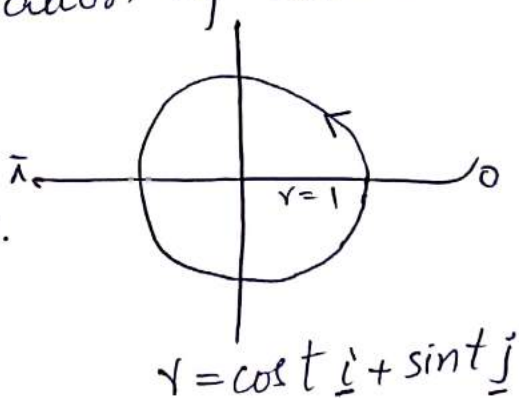
Now adding both equations

$$x^2 + y^2 = \cos^2 t + \sin^2 t$$

$$x^2 + y^2 = 1$$

So we got an equation of circle whose radius is 1.

in anti-clockwise direction.



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Example:- Sketch the graph of vector-valued function $r(t) = \cos t\, i + \sin t\, j + 2k \quad 0 \le t \le 2\pi$

As $r = r(t)$

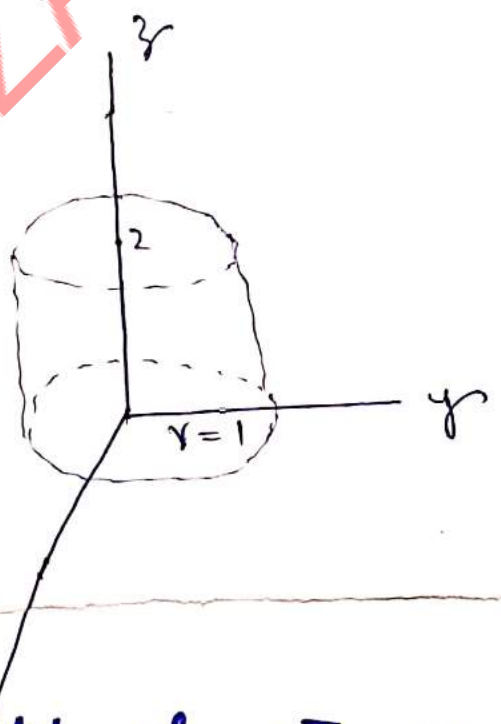
$r = xi + yj + zk$

$\cos t\, i + \sin t\, j + 2k = xi + yj + zk$

Comparing $x = \cos t$, $y = \sin t$, $z = 2$

squaring & adding first two equations

$x^2 + y^2 = 1$



stretch along z-axis.

$r = \cos t\, i + \sin t\, j + 2k$

Example:- Describe the graph of vector equ.

$r = (-2+t)i + 3tj + (5-4t)k$

Comparing $x = -2+t$, $y = 3t$, $z = 5-4t$

Graph is a line passing through $(-2, 0, 5)$ & parallel to $i + 3j - 4k$

(ignoring constants & t).

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Example:- The graph of vector-valued func $r(t) = t\underline{i} + t^2\underline{j} + t^3\underline{k}$ is called twisted cubic. Show That This curve ~~lies~~ Lies on parabolic cyllinder $y = x^2$ & sketch graph for $t \geq 0$. $r(t) = t\underline{i} + t^2\underline{j} + t^3\underline{k}$

Corresponding equations are:-

$$x = t, \quad y = t^2, \quad z = t^3$$

If we put $x = t$ in $y = t^2$ it becomes $y = x^2$ which is equation of parabolic cyllinder. So This curve Lies on parabolic cyllinder.

If we start t from origion ($t = 0$) by increasing "t" values of x, y, z also increases.

Graphs of constant vector-valued func.

If c is a constant vector in The sense That it does not depend on parameter, Then The graph of $r = c$ is a single point since The radius vector remains fixed with its tip at c.

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In 2d-space If $c = x_0 \underline{i} + y_0 \underline{j}$, then graph is the point (x_0, y_0) & if $c = x_0 \underline{i} + y_0 \underline{j} + z_0 k$ (3d-space) then graph is point (x_0, y_0, z_0).

x_0, y_0, z_0 are constant components.

Example:- The graph of $r(t) = \cos t \underline{i} + \sin t \underline{j} + 2k$

As $x = \cos t$ & $y = \sin t$ & $z = 2$

They form equ. of circle with radius 1 & lying in plane $z = 2$.

$$|r(t)| = \sqrt{x^2 + y^2 + z^2} = \sqrt{\cos^2 t + \sin^2 t + (2)^2}$$

$$= \sqrt{1 + 4} = \sqrt{5}$$

So the circle lies on surface of a sphere of radius $\sqrt{5}$.

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Lecture 28

Limits of vector valued functions

Limit of vector valued function is defined to be vector that results by taking limit of each component.

In 2d-Space $r(t) = x(t)\underline{i} + y(t)\underline{j}$ is defined as

$$\lim_{t\to\alpha} r(t) = \lim_{t\to\alpha} x(t)\underline{i} + \lim_{t\to\alpha} y(t)\underline{j}$$

In 3d-space $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$ is defined as

$$\lim_{t\to\alpha} r(t) = \lim_{t\to\alpha} x(t)\underline{i} + \lim_{t\to\alpha} y(t)\underline{j} + \lim_{t\to\alpha} z(t)\underline{k}$$

If lim of any component does not exist Then limit of $r(t)$ does not exist.

Same goes for $\lim_{t\to\alpha^+}$, $\lim_{t\to\alpha^-}$, $\lim_{t\to\infty^+}$, $\lim_{t\to\infty^-}$

We can write $\lim_{t\to\alpha} r(t) = L$

Length & direction of $r(t)$ approaches to length & direction of L as $t\to\alpha$.

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Continuity of vector valued functions

r is continuous at t_0 if

$r(t_0)$ is defined.

$\lim_{t \to t_0} r(t)$ exists.

$\lim_{t \to t_0} r(t) = r(t_0)$

The graph of a continuous vector-valued function is unbroken curve.

Derivatives of vector valued functions:-

The vector-valued function's derivative is

$$r'(t) = \lim_{h \to 0} \frac{r(t+h) - r(t)}{h}$$

$\frac{f(x+h) - f(x)}{h}$

Theorem:- If $r(t) = x(t)\underline{i} + y(t)\underline{j}$ in 2d.

Then $r'(t) = x'(t)\underline{i} + y'(t)\underline{j}$

Similarly if $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$ in 3d Then $r'(t) = x'(t)\underline{i} + y'(t)\underline{j} + z'(t)k$

Proof:-

As $r'(t) = \lim_{h \to 0} \frac{r(t+h) - r(t)}{h}$

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$$r'(t) = \lim_{h\to 0} \frac{[x(t+h)\underline{i} + y(t+h)\underline{j}] - [x(t)\underline{i} + y(t)\underline{j}]}{h}$$

$$= \lim_{h\to 0} \frac{[x(t+h) - x(t)]}{h}\underline{i} + \lim_{h\to 0}\left[\frac{y(t+h) - y(t)}{h}\right]\underline{j}$$

$$= x'(t)\underline{i} + y'(t)\underline{j}$$

Similarly for 3d-space.

Notations:-

$$\frac{d}{dt}r(t), \quad \frac{dr}{dt}, \quad r'(t), \quad r'$$

Example:- $r(t) = t^2\underline{i} + t^3\underline{j}$, $r'(t) = ?$, $r'(1) = ?$

$$r'(t) = \frac{d}{dt}(t^2)\underline{i} + \frac{d}{dt}(t^3)\underline{j}$$

$$r'(t) = 2t\underline{i} + 3t^2\underline{j}$$

now put $t = 1$

$$r'(1) = 2(1)\underline{i} + 3(1)^2\underline{j}$$

$$r'(1) = 2\underline{i} + 3\underline{j}$$

Tangent vectors & Tangent lines

geometrical interpretation of derivative

Let P be a point on graph of vector-valued func $r(t)$ & let $r(t_0)$ be radius vector from origin to P. If $r'(t_0)$ exists & $r'(t_0) \neq 0$ then we call

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$r'(t_0)$ The tangent vector to the graph of r at $r(t_0)$

Equation:- Vector equation of Tangent line is

$$r = r(t_0) + t\, r'(t_0)$$

Example:- Find parametric equ. of Tangent line to circular helix $x=\cos t$, $y=\sin t$, $z=t$ at point where $t_0 = \frac{\pi}{6}$.

Equating $r(t) = x\underline{i} + y\underline{j} + z\underline{k}$

$$r(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (t)k \longrightarrow (1)$$

$$r'(t) = (-\sin t)\underline{i} + (\cos t)\underline{j} + k \longrightarrow (2)$$

Put $t = \frac{\pi}{6}$ in (1) & (2) respectively

$$r\left(\frac{\pi}{6}\right) = \left(\cos\frac{\pi}{6}\right)\underline{i} + \left(\sin\frac{\pi}{6}\right)\underline{j} + \left(\frac{\pi}{6}\right)\underline{k}$$

$$r\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}k \quad \&$$

$$r'\left(\frac{\pi}{6}\right) = -\frac{1}{2}\underline{i} + \frac{\sqrt{3}}{2}\underline{j} + k$$

Now equ. of Tangent

$$r = r(t_0) + t\, r'(t_0)$$

$$= \left(\frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}\underline{k}\right) + t\left(\frac{-1}{2}\underline{i} + \frac{\sqrt{3}}{2}\underline{j} + k\right)$$

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$$= \left(\frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}\underline{k}\right) + -\frac{1}{2}t\,\underline{i} + \frac{\sqrt{3}}{2}t\,\underline{j} + tk$$

$$= \left(\frac{\sqrt{3}}{2} - \frac{1}{2}t\right)\underline{i} + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}t\right)\underline{j} + \left(\frac{\pi}{6} + t\right)\underline{k}$$

Comparing with $r = x\underline{i} + y\underline{j} + z\underline{k}$

$$x = \frac{\sqrt{3}}{2} - \frac{1}{2}t \;,\quad y = \frac{1}{2} + \frac{\sqrt{3}}{2}t \;,\quad z = \frac{\pi}{6} + t$$

Example:- The graph of $r(t) = t^2\underline{i} + t^3\underline{j}$ is called a semi-cubical parabola. Find a vector equation of tangent line to graph of $r(t)$ at

(a) The point $(0,0)$

(b) The point $(1,1)$

$$r(t) = t^2\underline{i} + t^3\underline{j}$$

$$r'(t) = 2t\underline{i} + 3t^2\underline{j}$$

(a) here $t=0$ so $r'(0) = 0$. There is no Tangent vector at the point & Tangent line does not exist.

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(b) here $t=1$ So vector equation will be

$$r = r(t_0) + t\,r'(t_0)$$

$$r = r(1) + t\,r'(1) \quad \text{—— (1)}$$

$$r(t) = t^2\underline{i} + t^3\underline{j}$$

So $r(1) = \underline{i} + \underline{j}$

& $r'(t) = 2t\underline{i} + 3t\underline{j}$

$r'(1) = 2\underline{i} + 3\underline{j}$

putting values in (1)

$$r = r(1) + t\,r'(1)$$

$$= (\underline{i} + \underline{j}) + t(2\underline{i} + 3\underline{j})$$

$$= \underline{i} + \underline{j} + 2t\underline{i} + 3t\underline{j}$$

$$= (1+2t)\underline{i} + (1+3t)\underline{j}$$

Smooth Function:-

$r(t)$ is called smooth function if components of r have continuous derivatives w.r.t t.

& $r'(t) \neq 0$ for any value of t.

Same for 2d & 3d- space.

A parametric eq will be called smooth if has a Tangent line at every point on its graph.

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Properties of derivatives:-

- $\frac{d}{dt}(c) = 0$
- $\frac{d}{dt}[k\, r(t)] = k \frac{d}{dt}[r(t)]$
- $\frac{d}{dt}[r_1(t) + r_2(t)] = \frac{d}{dt}[r_1(t)] + \frac{d}{dt}[r_2(t)]$
- $\frac{d}{dt}[r_1(t) - r_2(t)] = \frac{d}{dt}[r_1(t)] - \frac{d}{dt}[r_2(t)]$
- $\frac{d}{dt}[r_1(t) \cdot r_2(t)] = [r_1(t)]\frac{d}{dx}[r_2(t)] + [r_2(t)]\frac{d}{dx}[r_1(t)]$

Dot Product in 2d.

- $\frac{d}{dt}[r_1(t) \cdot r_2(t)] = r_1 \cdot \frac{dr_2}{dt} + \frac{dr_1}{dt} \cdot r_2$

Cross product in 3d.

- $\frac{d}{dt}[r_1(t) \times r_2(t)] = r_1 \times \frac{dr_2}{dt} + \frac{dr_1}{dt} \times r_2$

Order doesn't matter in 2d (Dot Product)
But matters in 3d (Cross Product)

Theorem:- If $r(t)$ is a vector-valued function in 2 space or 3 space & $|r(t)|$ is constant for all t, then $r(t) \cdot r'(t) = 0$ (orthogonal vectors)

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keeping $r_1(t) = r_2(t) = r(t)$ derivative of Dot product becomes

$$\frac{d}{dt}\left[r(t).r(t)\right] = r(t).\frac{dr}{dt} + \frac{dr}{dt}.r(t)$$

or

$$\frac{d}{dt}\left[|r(t)|\right]^2 = 2r(t).\frac{dr}{dt}.$$

But $\left[|r(t)|\right]^2$ is constant, so its derivative is zero. So

$$2r(t)\frac{dr}{dt} = 0$$

$$r(t).\frac{dr}{dt} = 0$$

(i-e) $r(t)$ is perpendicular to $\frac{dr}{dt}$.

Integrals of vector valued functions:-

(a) if $r(t) = x(t)\underline{i} + y(t)\underline{j}$, then

$$\int r(t)\,dt = \left(\int x(t)\,dt\right)\underline{i} + \left(\int y(t)\,dt\right)\underline{j}$$

$$\int_a^b r(t)\,dt = \left(\int_a^b x(t)\,dt\right)\underline{i} + \left(\int_a^b y(t)\,dt\right)\underline{j}$$

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If $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$

Then

$$\int r(t)\,dt = \left(\int x(t)\,dt\right)\underline{i} + \left(\int y(t)\,dt\right)\underline{j} + \left(\int z(t)\,dt\right)\underline{k}$$

$$\int_a^b r(t)\,dt = \left(\int_a^b x(t)\,dt\right)\underline{i} + \left(\int_a^b y(t)\,dt\right)\underline{j} + \left(\int_a^b z(t)\,dt\right)\underline{k}$$

Example: $r(t) = 2t\,\underline{i} + 3t^2\,\underline{j}$

(a) $\int r(t)\,dt$

$= \int (2t\,\underline{i} + 3t^2\,\underline{j})\,dt$

$= \left(\int 2t\,dt\right)\underline{i} + \left(\int 3t^2\,dt\right)\underline{j}$

$= \left(\frac{2t^2}{2} + C_1\right)\underline{i} + \left(\frac{3t^3}{3} + C_2\right)\underline{j}$

$= t^2\underline{i} + C_1\underline{i} + t^3\underline{j} + C_2\underline{j}$

$= t^2\underline{i} + t^3\underline{j} + C$ where $C_1\underline{i} + C_2\underline{j} = C$

(b) $\int_0^2 r(t)\,dt$

$= \left[t^2\right]_0^2\underline{i} + \left[t^3\right]_0^2\underline{j}$

$= (2^2 - 0^2)\underline{i} + (2^3 - 0^3)\underline{j}$

$= 4\underline{i} + 8\underline{j}$

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Properties of integral:-

$\frac{d}{dx}\left(\int x^2 dx\right)$

$$\int c\, r(t)\, dt = c \int r(t)\, dt$$

$$\int [r_1(t) + r_2(t)]\, dt = \int r_1(t)\, dt + \int r_2(t)\, dt$$

$$\int [r_1(t) - r_2(t)]\, dt = \int r_1(t)\, dt - \int r_2(t)\, dt$$

These properties also hold for definite integral.

In addition $\frac{d}{dt}\left(\int r(t)\, dt\right) = r(t)$

This shows indefinite integral of $r(t)$'s set of antiderivatives of $r(t)$ just as for real valued function.

$$\int r(t)\, dt = R(t) + C$$

$$\int_a^b r(t)\, dt = R(t)\Big]_a^b = R(b) - R(a).$$

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Lecture 29.

Change of Parameter

It is possible for different valued functions to have same graph. For example

$$r = (3\cos t)\hat{i} + 3(\sin t)\hat{j} \qquad 0 \le t \le 2\pi$$

&

$$r = \left[3\cos\left(\frac{s}{3}\right)\right]\hat{i} + \left[3\sin\left(\frac{s}{3}\right)\right]\hat{j} \qquad 0 \le s \le 6\pi$$

both the function have same graph ↑ circle of radius centered at origin. (here we made substitution $t = \frac{s}{3}$. real valued function). We can say that $r = g(u)$.

New vector valued function of u is smooth if original vector valued function of t is smooth.

If $\Rightarrow$ g is differentiable.

$\Rightarrow$ g' is continuous.

$\Rightarrow$ $g'(u) \neq 0$ for any u in domain of g.

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Arc length:-

⇒ If $x = x(t)$ & $y(t)$ in 2-space

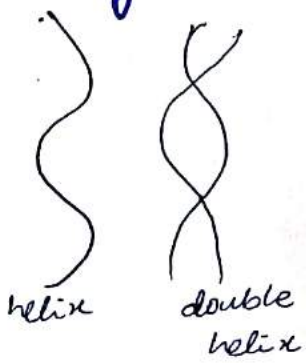
Then arc length $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

⇒ If $x = x(t)$, $y = y(t)$ & $z = z(t)$ is 3-space

arc length $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$

Example:- Find arc length of that portion of the circular helix $x = \cos t$, $y = \sin t$, $z = t$

Take $t = 0$ & $t = \pi$, $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$



$$L = \int_0^{\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2}\, dt$$

$$= \int_0^{\pi} \sqrt{\sin^2 t + \cos^2 t + 1}\, dt$$

$$= \int_0^{\pi} \sqrt{2}\, dt$$

$$= \sqrt{2} \int_0^{\pi} 1\, dt$$

$$= \sqrt{2}\,(t)\Big]_0^{\pi} \Rightarrow \sqrt{2}\,(\pi - 0)$$

$$L = \sqrt{2}\,\pi \quad \text{Ans.}$$

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Arc length as a parameter:-

The best parameter to use for representing a curve in 2-space or 3-space parametrically is length of arc measured along the curve from some reference point.

Steps:-

⇒ Select an arbitrary point on curve as reference point. (P)

⇒ Starting from reference point, choose one direction along curve ~~as~~ to be +ve & other to be -ve.

⇒ Let s be an arc Then s is +ve if point is +ve & ~~conversely~~ vice versa.

Theorem:- (a) Let C be a curve in 2-space parametrically by $x = x(t)$ & $y = y(t)$ where $x'(t)$ & $y'(t)$ are continuous functions. if S is arc length Then

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

(b) Let C be a curve in 3-space parametrically by $x = x(t)$, $y = y(t)$ & $z = z(t)$ where $x'(t)$, $y'(t)$ & $z'(t)$ are continuous, Then arc length

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$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}$$

Proof:- if $t > t_0$

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, dy\right.$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

Similarly for 3-d.

$\frac{ds}{dt}$ does not depend on t_0, its value is independent.

Example:- $x = 2t + 1$, $y = 3t - 2$

here reference point $(1, -2)$

replace t by u

$$x = 2u + 1, \quad y = 3u - 2$$

$$\frac{dx}{du} = 2, \quad \frac{dy}{du} = 3$$

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$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{(2)^2 + (3)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{13}\, du$$

$$= \sqrt{13} \int_{0}^{t} 1\, du \Rightarrow \sqrt{13}\, [u]_0^t$$

$$= \sqrt{13}(t-0) \Rightarrow \sqrt{13}\, t$$

So $s = \sqrt{13}\, t$

$$\frac{s}{\sqrt{13}} = t$$

put in

$$x = 2t + 1 \Rightarrow 2\left(\frac{s}{\sqrt{13}}\right) + 1 \Rightarrow \frac{2}{\sqrt{13}} s + 1$$

$$y = 3(t) - 2 \Rightarrow 3\left(\frac{s}{\sqrt{13}}\right) - 2 \Rightarrow \frac{3}{\sqrt{13}} s - 2$$

So These are parametric equations.

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Example:- Find parametric equations for the circle $x = a\cos t$, $y = a\sin t$ using arc length s as parameter, with reference point for s being $(a, 0)$, $a > 0$. $\quad 0 \le t \le 2\pi$

Replace t by u.

$$x = a\cos u, \quad y = a\sin u$$

$$\frac{dx}{du} = -a\sin u, \quad \frac{dy}{du} = a\cos u$$

Now
$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{(-a\sin u)^2 + (a\cos u)^2}\, du$$

$$= \int_{0}^{t} \sqrt{a^2\sin^2 u + a^2\cos^2 u}\, du$$

$$= \int_{0}^{t} \sqrt{a^2(\sin^2 u + \cos^2 u)}\, du$$

$$= \int_{0}^{t} \sqrt{a^2}\, du \Rightarrow \int_{0}^{t} a\, du \Rightarrow a\int_{0}^{t} du$$

$$= a[u]_0^t \Rightarrow a(t - 0) \Rightarrow at \quad \text{Ans.}$$

$$s = at, \quad \text{or} \quad t = \frac{s}{a}, \quad \text{So}$$

$$x = a\cos\left(\frac{s}{a}\right) \quad \text{and} \quad y = a\sin\left(\frac{s}{a}\right)$$

$0 \le t \le 2\pi$

$0 \le \frac{s}{a} \le 2\pi$

$0 \le s \le 2\pi a$

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Example:- Find arc length of curve

$$r(t) = t^3 \underline{i} + t \underline{j} + \frac{1}{2}\sqrt{6}\, t^2 \underline{k} \qquad 1 \le t \le 3$$

$$x = t^3, \quad y = t, \quad z = \frac{1}{2}\sqrt{6}\, t^2$$

$$\frac{dx}{dt} = 3t^2, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = \frac{1}{2}\sqrt{6}(2t) = \sqrt{6}\, t$$

$$\text{Arc length} = \int_1^3 \sqrt{(3t^2)^2 + (1)^2 + (\sqrt{6}\, t)^2}\, dt$$

$$= \int_1^3 \sqrt{9t^4 + 1 + 6t^2}\, dt$$

$$= \int_1^3 \sqrt{(3t^2)^2 + (1)^2 + 2(3t^2)(1)}\, dt$$

$$= \int_1^3 \sqrt{(3t^2+1)^2}\, dt$$

$$= \int_1^3 (3t^2+1)\, dt$$

$$= \left[\frac{3t^3}{3} + t\right]_1^3 \Rightarrow \left[t^3 + t\right]_1^3$$

$$= (3^3 + 3) - (1^3 + 1) \Rightarrow (27 + 3) - (1+1)$$

$$= 30 - 2 \Rightarrow 28 \quad \underline{\text{Ans}}.$$

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Lecture 30

Exact Differential

If $z = f(x,y)$, then $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

If $z = f(x,y,w)$, then $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy + \frac{\partial z}{\partial w}dw$

Exercise

1. $z = x^2 + y^2$, $dz = 2x\,dx + 2y\,dy \Rightarrow 2(x\,dx + y\,dy)$

2. $z = x^3 \sin 2y$, $dz = 3x^2 \cdot \sin 2y\,dx + x^3 \cos 2y \cdot 2\,dy$

$= x^2(3\sin 2y\,dx + 2x\cos 2y\,dy)$

3. $z = (2x-1)e^{3y}$, $dz = 2e^{3y}dx + (2x-1)e^{3y}\cdot 3dy$

$= e^{3y}(2dx + 3(2x-1)dy)$

4. $z = x^2 + 2y^2 + 3w^2$, $dz = 2x\,dx + 4y\,dy + 6w\,dw$

$= 2(x\,dx + 2y\,dy + 3w\,dw)$

5. $z = x^3y^2w$, $dz = 3x^2y^2w\,dx + x^3(2y)w\,dy + x^3y^2\,dw$

$= x^2y(3yw\,dx + 2xw\,dy + xy\,dw)$

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Exact differential:-

As in $z = f(x,y)$, $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

Work in reverse now

Any expression like $dz = Pdx + Qdy$ can be written as $P = \frac{\partial z}{\partial x}$ & $Q = \frac{\partial z}{\partial y}$

Now $\frac{\partial P}{\partial y} = \frac{\partial^2 z}{\partial y \partial x}$ & $\frac{\partial Q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y}$

Also $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

So dz is exact differential if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

Example:- $dz = (3x^2+4y^2)dx + 8xy\,dy$

Comparing with $Pdx + Qdy$

$P = 3x^2+4y^2$ & $Q = 8xy$

$\frac{\partial P}{\partial y} = 8y$ $\qquad$ $\frac{\partial Q}{\partial x} = 8y$

So dz is an exact differential.

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Example:- $dz = (1+8xy)\,dx + 5x^2\,dy$

here $P = 1+8xy$ & $Q = 5x^2$

$\frac{\partial P}{\partial y} = 8x$ $\qquad$ $\frac{\partial Q}{\partial x} = 10x$

Both are not equal so dz is not an exat differential.

Exercise:-

1. $dz = 4x^3y^3\,dx + 3x^4y^2\,dy$

$P = 4x^3y^3$ | $Q = 3x^4y^2$

$\frac{\partial P}{\partial y} = 4x^3(3y^2)$ | $\frac{\partial Q}{\partial x} = 3(4x^3)y^2$

$= 12x^3y^2$ | $= 12x^3y^2$

dz is exact differential.

2. $dz = (4x^3y + 2xy^3)\,dx + (x^4 + 3x^2y^2)\,dy$

$P = 4x^3y + 2xy^3$ | $Q = x^4 + 3x^2y^2$

$\frac{\partial P}{\partial y} = 4x^3(1) + 2x(3y^2)$ | $\frac{\partial Q}{\partial x} = 4x^3 + 3(2x)y^2$

$= 4x^3 + 6xy^2$ | $= 4x^3 + 6xy^2$

YES

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3. $dz = (15y^2e^{3x} + 2xy^2)dx + (10ye^{3x} + x^2y)dy$

$P = 15y^2e^{3x} + 2xy^2$

$\frac{\partial P}{\partial y} = 15(2y)e^{3x} + 2x(2y)$

$= \underline{30ye^{3x} + 4xy}$

$Q = 10ye^{3x} + x^2y$

$\frac{\partial Q}{\partial x} = 10ye^{3x} \cdot 3 + 2xy$

$= \underline{30ye^{3x} + 2xy}$

dz is not exact differential.

4. $dz = (3x^2e^{2y} - 2y^2e^{3x})dx + (2x^3e^{2y} - 2ye^{3x})dy$

$P = 3x^2e^{2y} - 2y^2e^{3x}$

$\frac{\partial P}{\partial y} = 3x^2e^{2y} \cdot 2 - 4y^2e^{3x}$

$= \underline{6x^2e^{2y} - 4y^2e^{3x}}$

NO

$Q = 2x^3e^{2y} - 2ye^{3x}$

$\frac{\partial Q}{\partial x} = 6xe^{2y} - 2ye^{3x} \cdot 3$

$= \underline{6x^2e^{2y} - 6ye^{3x}}$

5. $dz = (4y^3\cos 4x + 3x^2\cos 2y)dx + (3y^2\sin 4x - 2x^3\sin 2y)dy$

$P = 4y^3\cos 4x + 3x^2\cos 2y$

$\frac{\partial P}{\partial y} = 4(3y^2)\cos 4x + 3x^2(-\sin 2y) \cdot 2$

$= \underline{12y^2\cos 4x - 6x^2\sin 2y}$

$Q = 3y^2\sin 4x - 2x^3\sin 2y$

$\frac{\partial Q}{\partial x} = 3y^2(\cos 4x) \cdot 4 - 2(3x^2)\sin 2y$

$= \underline{12y^2\cos 4x - 6x^2\sin 2y}$

YES

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Integration of Exact differentials

$dz = P\,dx + Q\,dy$ where $P = \frac{\partial z}{\partial x}$ & $Q = \frac{\partial z}{\partial y}$

$\therefore z = \int P\,dx$ & $z = \int Q\,dy$

Example:- $dz = (2xy + 6x)\,dx + (x^2 + 2y^3)\,dy$

$P = 2xy + 6x$, $Q = x^2 + 2y^3$

$z = \int P\,dx$

$= \int (2xy + 6x)\,dx$

$= 2y \cdot \frac{x^2}{2} + 6\frac{x^2}{2} + F(y)$

$z = x^2y + 3x^2 + F(y)$

$z = \int Q\,dy$

$= \int (x^2 + 2y^3)\,dy$

$= x^2y + 2\frac{y^4}{4} + F(x)$

$z = x^2y + \frac{y^4}{2} + F(x)$

To make them equal $F(y)$ should be $\frac{y^4}{2}$ & $F(x)$ should be $3x^2$

So $z = x^2y + 3x^2 + \frac{y^4}{2}$

Example: Integrate $dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$. to make

$z = 2e^{4x} + x^2y^2 + \sin 4y$

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$$dz = (8e^{4x} + 2xy^2)dx + (4\cos 4y + 2x^2y)dy$$

here

$$P = 8e^{4x} + 2xy^2$$

$$z = \int P\,dx$$

$$= \int (8e^{4x} + 2xy^2)\,dx$$

$$= \frac{8e^{4x}}{4} + 2\frac{x^2}{2}\cdot y^2 + F(y)$$

$$= 2e^{4x} + x^2y^2 + F(y)$$

$$Q = 4\cos 4y + 2x^2y$$

$$z = \int Q\,dy$$

$$z = \int (4\cos 4y + 2x^2y)\,dy$$

$$= \frac{4(\sin 4y)}{4} + 2x^2\frac{y^2}{2} + f(x)$$

$$= \sin 4y + x^2y^2 + F(x)$$

here $F(y) = \sin 4y$ & $F(x) = 2e^{4x}$

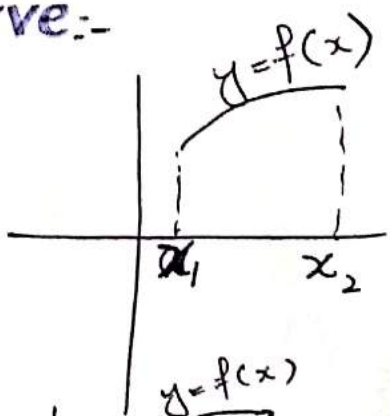
So

$$z = 2e^{4x} + x^2y^2 + \sin 4y.$$

Area enclosed by closed curve:-

Remember we used integration to find area of a plane.

$$A_1 = \int_{x_1}^{x_2} y\,dx = \int_{x_1}^{x_2} f(x)\,dx$$

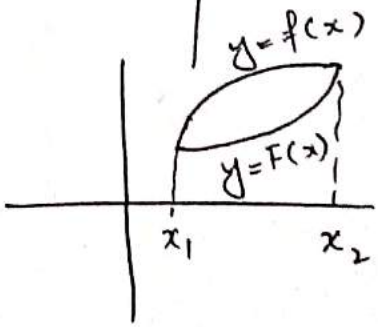


If there are two curves

$$A = A_1 - A_2$$

$$A = \int_{x_1}^{x_2} f(x)\,dx - \int_{x_1}^{x_2} F(x)\,dx$$

$$= -\left\{ \int_{x_1}^{x_2} f(x)\,dx + \int_{x_1}^{x_2} F(x)\,dx \right.$$



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Final result can also be written as

$$A = -\oint y\,dy$$

$\oint$ indicates that integral is to be ~~written~~ evaluated round closed boundry in the +ve.

Example:- Determine area enclosed by

The graph of $y = x^3$ & $y = 4x$ for $x \geq 0$.

$Y = x^3$ & $y = 4x$

To find limits

$$x^3 = 4x$$
$$x^3 - 4x = 0$$
$$x(x^2 - 4) = 0$$

$x = 0$

$x^2 - 4 = 0$

$x^2 = 4$

$\sqrt{x^2} = \sqrt{(2)^2}$

$x = \pm 2$

$x = 2 \quad \because x \geq 0$

in anticlockwise direction

C_1 : $y = x^3$, limit $x = 0$ & $x = 2$

C_2 : $y = 4x$, limit $x = 2$ & $x = 0$

$$A = -\oint y\,dy \Rightarrow -\left\{ \int_0^2 x^3\,dx + \int_2^0 4x\,dx \right\}$$

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$$= -\left\{ \left[\frac{x^4}{4}\right]_0^2 + \left[4\frac{x^2}{2}\right]_2^0 \right\}$$

$$= -\left(\left[\frac{(2)^4}{4} - \frac{(0)^4}{2}\right] + \left[2(0)^2 - 2(2)^2\right] \right)$$

$$= -\left(\frac{16}{4} - 8\right)$$

$$= -(4-8) \Rightarrow -(-4) \Rightarrow 4 \quad \underline{\text{Ans.}}$$

Example:- Find area of Triangle with vertices $O(0,0)$ $A(5,3)$ & $B(2,6)$.

m = Slope of $OA = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-0}{5-0} = \frac{3}{5}$

m = Slope of $BA = \frac{6-3}{2-5} = \frac{3}{-3} = -1$

m = Slope of $OB = \frac{6-0}{2-0} = \frac{6}{2} = 3$

eq. of $OA \Rightarrow y - y_1 = m(x - x_1)$

$$y - 0 = \frac{3}{5}(x-0)$$

$$y = \frac{3}{5}x$$

eq. of $BA \Rightarrow y - 5 = -1(x-3)$

$$y = -x + 3 + 5$$

$$y = -x + 8$$

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eq. of OB ⇒ $y-0 = 3(x-0)$

$y = 3x$

Now

$$A = -\oint y\,dy$$

$$= -\left(\int_0^5 \frac{3}{5}x\,dx + \int_5^2 (-x+8)\,dy + \int_2^0 3x\,dx \right)$$

$$= -\left(\frac{3}{5}\left[\frac{x^2}{2}\right]_0^5 + \left[\frac{-x^2}{2}+8x\right]_5^2 + \left[\frac{3x^2}{2}\right]_2^0 \right)$$

$$= -\left(\frac{3}{5}\left[\frac{5^2-0^2}{2}\right] + \left[\left(\frac{-(2)^2}{2}+8(2)\right) - \left(\frac{-(5)^2}{2}+8(5)\right)\right] + \left[\quad\right] \right)$$

$$\left[\frac{3(0)^2}{2} - \frac{3(2)^2}{2}\right]$$

$$= -\left[\frac{3}{5}\left(\frac{25}{2}\right) + \left(14 - \frac{55}{2}\right) + (0-6)\right]$$

$$= -\left(\frac{15}{2} + 14 - \frac{55}{2} - 6\right)$$

$$= -\left(\frac{15+28-55-12}{2}\right) \Rightarrow -\left(\frac{-24}{2}\right)$$

$\Rightarrow$ 12 Ans.

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Line integral:-

If a field exists in xy-plane, producing a Force F on a particle at K. Then F can be divided into two components.

F_1 along Tangent & F_a along normal.

Approximate workdone in moving particle is $F_t \delta s$ if δs is the distance covered.

$$W = \sum F_t \delta s$$

$$= \lim_{\delta x \to 0} \sum F_t \delta x = \int F_t \, ds$$

or $$\int_{AB} F_t \, ds = \int_C F_t \, ds$$

where C is curve $y = f(x)$ between $A(x_1, y_1)$ & $B(x_2, y_2)$.

Alternative form:-

If F_t has a component P in x-direction & Q in y-direction then workdone is

$$P\delta x + Q\delta y$$

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So $\int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q\,dy)$

P & Q are functions of x & y.

In general.

$$\int_C F_t \, ds = \int_C (P\,dx + Q\,dy) \quad \left(\text{where } c \text{ is the curve}\right)$$

Lecture 31 Line Integral.

Example:- Evaluate $\int_C (x+3y)\,dx$ from $A(0,1)$ to $B(2,5)$ along the curve $y = 1+x^2$

In $\int_C (x+3y)\,dx$, $P = x+3y$ & $Q = 0$ & $c = 1+x^2$

$$\int_C (x+3(1+x^2))\,dx = \int_0^2 (x+3+3x^2)\,dx$$

$$= \left[\frac{x^2}{2} + 3x + \frac{3x^3}{3}\right]_0^2 = \left[\frac{2^2}{2} + 3(2) + (2)^3\right] - 0$$

$$= (2+6+8) \Rightarrow 16 \quad \text{Ans.}$$

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Example 2:- Evaluate $I = \int_C (x^2+y)dx + (x-y^2)dy$ from $A(0,2)$ to $B(2,5)$ along curve $y = 2+x$

$y = 2+x$

$dy = dx$

$P = x^2+y \Rightarrow x^2+2+x$

$Q = x-y^2 \Rightarrow x-(2+x)^2 \Rightarrow x-4-x^2-4x$

$= -3x-4-x^2 \Rightarrow -(3x+4+x^2)$

So

$$\int_C (P\,dx + Q\,dy)$$

$$= \int_0^2 \left[(x^2+x+2)\frac{dx}{} + (-(3x+4+x^2))\,dx \right]$$

$$= \int_0^2 \left[x^2+x+2-3x-4-x^2 \right] dx$$

$$= \int_0^2 (-2x-2)\,dx \Rightarrow \left[-\frac{2x^2}{2} - 2x \right]_0^2$$

$$= \left(-(2)^2 - 2(2) \right) - \left(-0^2 - 2(0) \right)$$

$$= (-4-4) - 0$$

$$= -8 \quad \text{Ans.}$$

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Example Evaluate $I = \int_C (x^2+2y)dx + xy\,dy$ from $(0,0)$ To $B(1,4)$ along curve $y = 4x^2$

$y = 4x^2$
$dy = 8x\,dx$

$P = x^2+2y \Rightarrow x^2+2(4x^2)$
$= x^2+8x^2 \Rightarrow 9x^2$

$Q = xy \Rightarrow x(4x^2) \Rightarrow 4x^3$

So
$$\int_C P\,dx + Q\,dy$$
$$= \int_0^1 9x^2\,dx + 4x^3(8x\,dx)$$
$$= \int_0^1 (9x^2 + 32x^4)\,dx \Rightarrow \left[\frac{9x^3}{3} + 32\frac{x^5}{5}\right]_0^1$$
$$\Rightarrow \left(\frac{9(1)^3}{3} + 32\frac{(1)^5}{5}\right) - \left(\frac{9(0)^3}{3} + 32\frac{(0)^5}{5}\right)$$
$$= \left(\frac{9}{3} + \frac{32}{5}\right) - 0 \Rightarrow 3 + \frac{32}{5} \Rightarrow \frac{15+32}{5}$$
$$= \frac{47}{5} \Rightarrow 9.4 \quad \text{Ans.}$$

Example:- Evaluate $\int_C (x^2+2y)dx + xy\,dy$ from $O(0,0)$ to $A(1,0)$ along line $y=0$ & Then from $A(1,0)$ To $B(1,4)$ along line $x=1$.

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(i) for line $y=0$ & $O(0,0)$, $A(1,0)$

$P = x^2 + 2y \Rightarrow x^2 + 2(0) = x^2$

$Q = xy \Rightarrow x(0) \Rightarrow 0$

$y = 0$

$dy = 0$

So

$$I_{OA} = \int_C P\,dx + Q\,dy$$

$$= \int_0^1 x^2\,dx + 0 \Rightarrow \int_0^1 x^2\,dx \Rightarrow \left[\frac{x^3}{3}\right]_0^1$$

$$= \left(\frac{(1)^3}{3} - \frac{0^3}{3}\right) \Rightarrow \frac{1}{3}$$

(ii) for line $x=1$ & $A(1,0)$ To $B(1,4)$

$P = x^2 + 2y \Rightarrow 1^2 + 2y \Rightarrow 1 + 2y$

$Q = xy = (1)y \Rightarrow y$

$x = 1$

$dx = 0$

So

$$I_{AB} = \int_C P\,dx + Q\,dy \Rightarrow \int_0^4 (1+2y)(0) + (y)\,dy$$

$$= \int_0^4 y\,dy \Rightarrow \left[\frac{y^2}{2}\right]_0^4 \Rightarrow \frac{(4)^2}{2} \Rightarrow \frac{16}{2} \Rightarrow 8$$

Then $I = I_{OA} + I_{AB}$

$$= \frac{1}{3} + 8 \Rightarrow \frac{1+24}{3} \Rightarrow \frac{25}{3} \text{ or } 8\frac{1}{3}$$ Ans.

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Properties of line integral:-

$\Rightarrow \int_C F\,ds = \int_c (P\,dx + Q\,dy)$

$\Rightarrow \int_{AB} F\,ds = -\int_{BA} F\,ds$

or

$\int_{AB} (P\,dx + Q\,dy) = -\int_{BA} (P\,dx + Q\,dy).$

Sign is reversed when direction is reversed.

$\Rightarrow$ For a path parallel to x-axis i.e $x = k$, $dx = 0$

So $\int_C P\,dx = 0 \quad \therefore \int_C Q\,dy$

$\Rightarrow$ For a path parallel to y-axis i.e $y = k$, $dy = 0$

$\int_C Q\,dy = 0 \quad \therefore \int_C P\,dx$

$\Rightarrow$ If path is not single valued, then it is divided into two sections. $y = f_1(x)$ A to K & $y = f_2(x)$ from K to B.

$\Rightarrow$ If path $\bar{AB}$ is divided into two ~~paths~~ parts AK & BK then $I_C = I_{AB} = I_{AK} + I_{KB}$

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$\Rightarrow$ In all cases actual path must be continuous & single valued.

Example:- Evaluate $I = \int_C (x+y)\,dx$ from $A(0,1)$ to $B(0,-1)$ along semi-circle $x^2+y^2=1$ for $x \geq 0$

here path is not single valued

$$x^2+y^2=1, \quad y^2 = 1-x^2$$

$$y = \pm\sqrt{1-x^2}$$

So (i) $y = \sqrt{1-x^2}$ from A to K

(ii) $y = -\sqrt{1-x^2}$ from K to B.

As $I = \int_C (x+y)\,dx$ here $Q = 0$

$P = x+y \Rightarrow x+\sqrt{1-x^2}$

So $I = \int_C (P\,dx + Q\,dy)$

will be $I = \int_C P\,dx$

$$I_{AB} = I_{AK} + I_{KB}$$

$$= \int_0^1 (x+\sqrt{1-x^2})\,dx + \int_1^0 (x-\sqrt{1-x^2})\,dx$$

$$= \int_0^1 (x+\sqrt{1-x^2})\,dx - \int_0^1 (x-\sqrt{1-x^2})\,dx$$

$$= \int_0^1 (x+\sqrt{1-x^2} - x + \sqrt{1-x^2})\,dx$$

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$$= \int_0^1 2\sqrt{1-x^2}\,dx$$

$$= 2\int_0^1 \sqrt{1-x^2}\,dx$$

If we put $x = \sin\theta$

$dx = \cos\theta\, d\theta$

put $x = 0$ in $x = \sin\theta$

$0 = \sin\theta$

$\theta = 0$

put $x = 1$

$1 = \sin\theta$

$\sin^{-1}(1) = \theta$

$\frac{\pi}{2} = \theta$

$$= 2\int_0^{\pi/2} \sqrt{1-\sin^2\theta}\,\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \cos\theta\cdot\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$\because \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}$

$$= \int_0^{\pi/2} (1+\cos 2\theta)\,d\theta \Rightarrow$$

$$= \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}$$

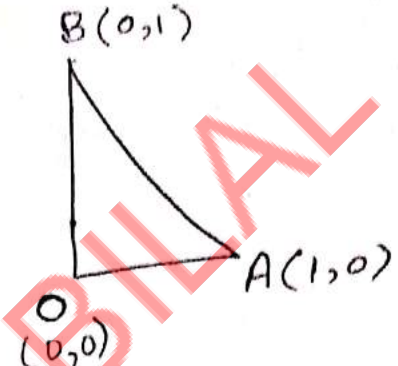
$$= \left(\frac{\pi}{2} + \frac{\sin\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right)$$

$$= \left(\frac{\pi}{2} + 0\right) - 0 \Rightarrow \frac{\pi}{2}$$ Ans.

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Example:- Evaluate $\oint (x^2 dx - 2xy\, dy)$ where c comprises three sides of triangle joining O(0,0) A(1,0) & B(0,1).



For OA, $y = 0$

$dy = 0$

$P = x^2$, $Q = 2xy \Rightarrow 2x(0) \Rightarrow 0$

So

$$I_1 = \int_{OA} x^2 dx \Rightarrow \int_0^1 x^2 dx \Rightarrow \left| \frac{x^3}{3} \right|_0^1$$

$$I_1 = \frac{(1)^3}{3} - \frac{(0)^3}{3} \Rightarrow \frac{1}{3} - 0 \Rightarrow \boxed{\frac{1}{3}}$$

For AB, $y = 1 - x$

$dy = -1\, dx$

$P = x^2$, $Q = -2xy$

$P = x^2$, $Q = -2x(1-x) \Rightarrow -2x + 2x^2$

So

$$I_2 = \int_{AB} x^2 dx + (-2x + 2x^2) - dx$$

$$\int_1^0 (x^2 + 2x - 2x^2)\, dx$$

$y - y_0 = m(x - x_0)$

$y - 0 = -1(x - 1)$

$y = -x + 1$

$y = 1 - x$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-1}{1-0} = -1$

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$$I_2 = \int_1^0 (-x^2 + 2x)\,dx$$

$$= \left[-\frac{x^3}{3} + \frac{2x^2}{2} \right]_1^0 \Rightarrow 0 - \left(-\frac{1^3}{3} + (1)^2 \right)$$

$$I_2 = -\left(-\frac{1}{3} + 1 \right) \Rightarrow -\left(\frac{-1+3}{3} \right) \Rightarrow \boxed{-\frac{2}{3}}$$

For BO, line $x = 0$

$dx = 0$

$P = x^2$, $Q = -2xy$

$P = 0$, $Q = 0$

$$I_3 \int_{BO} 0\,dy \Rightarrow 0$$

Finally $I = I_1 + I_2 + I_3$

$$= \frac{1}{3} - \frac{2}{3} + 0$$

$$= \frac{1-2}{3} \Rightarrow \boxed{\frac{-1}{3}}$$ Ans.

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Example:- Evaluate $\oint_c y\,dx$ when c is the circle $x^2+y^2=4$.

$$y^2 = 4 - x^2$$

$$y = \pm\sqrt{4-x^2}$$

$r^2 = 4$

$x = r = \pm 2$

y is not single valued so

$$I = \int_2^{-2} \sqrt{4-x^2}\,dx + \int_{-2}^{2} -\sqrt{4-x^2}\,dx$$

$$= \int_{+2}^{-2} \sqrt{4-x^2}\,dx + \int_{2}^{-2} +\sqrt{4-x^2}\,dx$$

$$= \int_{2}^{-2} 2\sqrt{4-x^2}\,dx$$

$$= 2\int_{-2}^{2} \sqrt{4-x^2}\,dx$$

$$= 2\times 2\int_{0}^{2} \sqrt{4-x^2}\,dx$$

$$= 4\int_{0}^{2} \sqrt{4-x^2}\,dx$$

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put $x = 2\sin\theta$,

$dx = 2\cos\theta\, d\theta$

Limits

$x = 0$	$x = 2$
$x = 2\sin\theta$	$2 = 2\sin\theta$
$0 = 2\sin\theta$	$1 = \sin\theta$
$0 = \sin\theta$	$\sin^{-1}(1) = \theta$
$\sin^{-1}(0) = \theta$	$\frac{\pi}{2} = \theta$
$0 = \theta$	

Now

$$-4\int_0^{\pi/2} \sqrt{4-(2\sin\theta)^2}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} \sqrt{4-4\sin^2\theta}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} \sqrt{4(1-\sin^2\theta)}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} 2\cos\theta\, 2\cos\theta\, d\theta$$

$$= -4\times 2\int_0^{\pi/2} 2\cos^2\theta\, d\theta \qquad \because 2\cos^2\frac{\theta}{2} = 1+\cos\theta$$

$$= -16\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= -8\int_0^{\pi/2} (1+\cos 2\theta)\, d\theta \Rightarrow -8\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}$$

$$= -8\left[\frac{\pi}{2} + \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right] \Rightarrow -8\left[\frac{\pi}{2}+0\right] \Rightarrow -8\left(\frac{\pi}{2}\right)$$

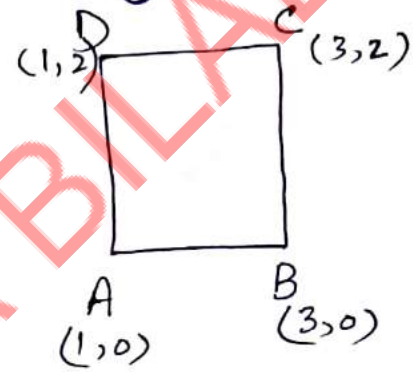
$$= -4\pi \quad \underline{\text{Ans.}}$$

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Lecture 32. Examples

Evaluate $I = \oint \{xy\,dx + (1+y^2)\,dy\}$ where c is boundry of rectangle joining A(1,0), B(3,0), C(3,2), D(1,2).



I_1 = line $y=0$, $dy=0$

$I_{AB} = I_1 = 0$

$I_2 = I_{BC}$ line $x=3$, $dx=0$

$$I_2 = \int_0^2 (1+y^2)\,dy \Rightarrow \left[y + \frac{y^3}{3}\right]_0^2 \Rightarrow \left(2 + \frac{2^3}{3}\right) - 0$$

$$= 2 + \frac{8}{3} \Rightarrow \frac{6+8}{3} = \frac{14}{3}$$

$I_3 = I_{CD}$ $y=2$, $dy=0$

$$I_3 = \int_3^1 x(2)\,dx \Rightarrow \left[2\,\frac{x^2}{2}\right]_3^1 = \left[x^2\right]_3^1$$

$$= (1)^2 - (3)^2 \Rightarrow 1 - 9 \Rightarrow -8$$

$I_4 = I_{DA}$ line $x=1$, $dx=0$

$$I_4 = \int_2^0 (1+y^2)\,dy \Rightarrow \left[y + \frac{y^3}{3}\right]_2^0 \Rightarrow 0 - \left(2 + \frac{2^3}{3}\right)$$

$$\Rightarrow \frac{-14}{3}$$ Ans

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Now $I = I_1 + I_2 + I_3 + I_4$

$= 0 + \frac{14}{3} - 8 - \frac{14}{3} \Rightarrow -8$ Ans.

Line integral w.r.t arc length

As $I = \int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q\,dy)$

if $I = \int_c f(x,y)\,ds = \int_c f(x,y) \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

where $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Example:- Evaluate $I = \int_c (4x + 3xy)\,ds$ where c is straight line joining O(0,0) to A(1,2). c is line $y = 2x$

$dy = 2$

$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$= \sqrt{1 + (2)^2} \Rightarrow \sqrt{5}$

$y - y_0 = m(x - x_0)$

$y - 0 = 2(x - 0)$

$y = 2x$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{2-0}{1-0}$

$= 2$

Now

$\int_0^1 (4x + 3xy)\sqrt{5}\,dx$

$\int_0^1 (4x + 3x(2x))\sqrt{5}\,dx$

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$$\int_0^1 (4x + 6x^2)\sqrt{5}\, dx$$

$$= \sqrt{5}\left[\frac{4x^2}{2} + \frac{6x^3}{3}\right]_0^1$$

$$= \sqrt{5}\left[2x^2 + 2x^3\right]_0^1$$

$$= \sqrt{5} \times 2\left[(1^2 + 1^3) - (0)\right]$$

$$= 2\sqrt{5}\,(2) \Rightarrow 4\sqrt{5} \quad \text{Ans.}$$

Parametric Equations:-

When x & y are expressed in parametric form $x = y(t)$ & $y = g(t)$, Then

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$I = \int_c f(x,y)\, ds = \int_{t_1}^{t_2} f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \frac{dy}{(dt)^2}}\, dt$$

Example:- Evaluate $I = \oint 4xy\, ds$, where c is defined as curve $x = \sin t$, $y = \cos t$. between $t = 0$ & $t = \frac{\pi}{4}$.

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$x = \sin t$, $y = \cos t$

$\frac{dx}{dt} = \cos t$, $\frac{dy}{dt} = -\sin t$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \Rightarrow \sqrt{\cos^2 t + (-\sin t)^2}$$

$$= \sqrt{1} = 1$$

$$I = \int_{t_1}^{t_2} f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^{\pi/4} 4xy\,(1)\,dt \Rightarrow \int_0^{\pi/4} 4 \sin t \cos t\, dt$$

$$= 2\int_0^{\pi/4} 2\sin t \cos t\, dt \Rightarrow 2\int_0^{\pi/4} \sin 2t\, dt$$

$$= +2\left[-\frac{\cos 2t}{2}\right]_0^{\frac{\pi}{4}} \Rightarrow -\left[\cos 2t\right]_0^{\frac{\pi}{4}}$$

$$= -\left[\cos 2\left(\frac{\pi}{4}\right) - \cos 2(0)\right]$$

$$= -\left(\cos\frac{\pi}{2} - \cos 0\right)$$

$$= -(0 - 1) \Rightarrow -(-1) \Rightarrow 1 \quad \underline{\text{Ans.}}$$

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Dependence of linear integral on path of integration.

Example:- Evaluate $I = \oint \{3x^2y^2dx + 2x^3y\,dy\}$

b/w $O(0,0)$ & $A(2,4)$

(a) along c_1 (i-e) $y = x^2$

(b) along c_2 (ie) $y = 2x$

(c) along c_3 (i-e) $x = 0$ from $(0,0)$ to $(0,4)$

& $y = 4$ from $(0,4)$ to $(2,4)$

(a)

$$I_1 = \int_c (3x^2y^2dx + 2x^3y\,dy) \quad \text{line} \quad y = x^2, \; dy = 2x\,dx$$

$$= \int_0^2 \left(3x^6dx + 2x^3(x^2)(2x)\,dx\right)$$

$$= \int_0^2 (3x^6 + 4x^6)\,dx \Rightarrow \int_0^2 7x^6\,dx$$

$$= 7\left[\frac{x^7}{7}\right]_0^2 \Rightarrow [x^7]_0^2 \Rightarrow (2)^7 - (0)^7 \Rightarrow 128$$

$$\boxed{I_1 = 128}$$

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(b) line $y = 2x$

$dy = 2dx$

$$I_2 = \int_C (3x^2y^2dx + 2x^3y\,dy)$$

$$= \int_0^2 \left[3x^2(2x)^2dx + 2x^3(2x)(2dx) \right]$$

$$= \int_0^2 \left[12x^4 + 8x^4 \right]dx \Rightarrow \int_0^2 20x^4\,dx$$

$$= 20\left[\frac{x^5}{5}\right]_0^2 \Rightarrow 4\left[x^5\right]_0^2 \Rightarrow 4\left[2^5 - 0^5\right]$$

$$= 4(32) \Rightarrow 128$$

(c) $I_{3a} = 0$

line $x = 0$ from (0,0) to (0,4)

$dx = 0$

$$I_{3b} = \int_C 3x^2(4)^2dx$$

line $y = 4$ from (0,4) to (2,4)

$dy = 0$

$$= \int_0^2 48x^2dx$$

$$= 48\left[\frac{x^3}{3}\right]_0^2 \Rightarrow 16\left[x^3\right]_0^2 \Rightarrow 16\left[2^3 - 0^3\right]$$

$$= 16(8) \Rightarrow 128 \quad \text{Ans.}$$

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Lecture 33 Examples

Evaluate $I = \int_c \{3y\,dx + (3x+2y)\,dy\}$ from $A(1,2)$ to $B(3,5)$.

If no path is given, then integrand is an exact differential of some function $z = f(x,y)$. As

$$I = \int_c \{P\,dx + Q\,dy\}$$

here $P = 3y$, $z = \int 3y\,dx = 3xy + f(y)$

$Q = 3x+2y$, $z = \int (3x+2y)\,dy = 3xy + y^2 + f(x)$

in this case $f(y) = y^2$ & $f(x) = 0$

hence $z = 3xy + y^2$

$$I = \int_{(1,2)}^{(3,5)} d(3xy+y^2) \Rightarrow \left[3xy+y^2\right]_{(1,2)}^{(3,5)}$$

$$= \left[3(3)(5)+(5)^2\right] - \left[3(1)(2)+(2)^2\right]$$

$$= 3\left[45+25\right] - \left[6+4\right] \Rightarrow 70 - 10$$

$$= 60 \quad \text{Ans.}$$

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Example:-

Evaluate $I = \int_C \{(x^2+ye^x)dx + (e^x+y)dy\}$ b/w

A(0,1) & B(1,2).

$P = x^2+ye^x$, $z = \int (x^2+ye^x)dx$

$= \frac{x^3}{3} + ye^x + f(y)$

$Q = (e^x+y)$, $z = \int Q\,dy \Rightarrow \int (e^x+y)\,dy$

$= e^x y + \frac{y^2}{2} + f(x)$

here $f(y) = \frac{y^2}{2}$ & $f(x) = \frac{x^3}{3}$

So $z = \frac{x^3}{3} + ye^x + \frac{y^2}{2}$

Then $I = \left[\frac{x^3}{3} + ye^x + \frac{y^2}{2}\right]_{(0,1)}^{(1,2)}$

$= \left(\frac{1}{3} + (2)e^1 + \frac{2^2}{2}\right) - \left(0 + (1)e^0 + \frac{1^2}{2}\right)$

$= \left(\frac{1}{3} + 2e + 2\right) - \left(1 + \frac{1}{2}\right)$

$= \frac{1+6e+6}{3} - \frac{3}{2} \Rightarrow \frac{12e+14-9}{6}$

$= \frac{12e+5}{6} \Rightarrow 2e + \frac{5}{6}$ Ans.

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If $(P dx + Q dy)$ is exact ~~function~~ differential

(a) $I = \int_c (P dx + Q dy)$ is independent of path.

(b) $I = \oint_c (P dx + Q dy)$ is zero.

Exact differentials in 3 independent variables:-

If $z = f(x, y, w)$ Then its exact differentia

will be $dz = P dx + Q dy + R dw$

& $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$; $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$; $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

if test is successful Then

(a) $\int_c (P dx + Q dy + R dw)$ is independent of path.

(b) $\oint_c (P dx + Q dy + R dw)$ is zero.

Example:- Verify that

$dz = (3x^2yw + 6x) dx + (x^3w - 8y) dy + (x^3y + 1) dw$

is in exact differential & hence evaluate

$\int_c dz$ from $A(1, 2, 4)$ to $B(2, 1, 3)$.

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$P = 3x^2yw + 6x$, $Q = x^3w - 8y$, $R = x^3y + 1$

$\frac{\partial P}{\partial y} = 3x^2w$, $\frac{\partial Q}{\partial x} = 3x^2w$, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$\frac{\partial P}{\partial w} = 3x^2y$, $\frac{\partial R}{\partial x} = 3x^2y$, $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$

$\frac{\partial Q}{\partial w} = x^3$, $\frac{\partial R}{\partial y} = x^3$, $\frac{\partial Q}{\partial w} = \frac{\partial R}{\partial y}$

So dz is exact differential.

To find z,

$z = \int P\,dx = \int (3x^2yw + 6x)\,dx \Rightarrow \frac{3x^3}{3}yw + \frac{6x^2}{2}$

$= x^3yw + 3x^2 + f(y) + F(w)$

$z = \int Q\,dy = \int (x^3w - 8y)\,dy \Rightarrow x^3yw - \frac{8y^2}{2}$

$= x^3yw - 4y^2 + F(w) + g(x)$

$z = \int R\,dy = \int (x^3y + 1)\,dw \Rightarrow x^3yw + w + f(y) + g(x)$

Now

$z = x^3yw + 3x^2 - 4y^2 + w$

$I = [x^3yw + 3x^2 - 4y^2 + w]$

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$$I = \left[x^3yw + 3x^2 - 4y^2 + w\right]_{(1,2,4)}^{(2,1,3)}$$

$$= \left[(2)^3(1)(3) + 3(2)^2 - 4(1)^2 + 3\right] - \left[(1)^3(2)(4) + 3(1)^2 - 4(2)^2 + 4\right]$$

$$= (24 + 12 - 4 + 3) - (8 + 3 - 16 + 4)$$

$$= 36 \quad \underline{\text{Ans.}}$$

Green's Theorem:- Let P & Q are two functions of x & y that are finite & continuous. If first partial derivatives are continuous, then

$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = -\oint_C (P\,dx + Q\,dy)$$

Example:- Evaluate

$I = \oint \{(2x - y)dx + (2y + x)dy\}$ around boundry C, the ellipse $x^2 + 9y^2 = 16$.

$P = 2x - y, \quad Q = 2y + x$

$\frac{\partial P}{\partial y} = 0 - 1 \Rightarrow -1, \quad \frac{\partial P}{\partial x} = 0 + 1 \Rightarrow 1$

$$I = -\iint_R (-1 - 1)\,dx\,dy = -\iint_R (-2)\,dx\,dy$$

$$= 2\iint_R dx\,dy$$ (Find area of ellipse & multip by 2 now)

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$x^2 + 9y^2 = 16$

divide by 16

$\frac{x^2}{16} + \frac{9y^2}{16} = 1$ comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{x^2}{(4)^2} + \frac{y^2}{\frac{(4)^2}{(3)^2}} = 1$

$a = 4, \quad b = \frac{4}{3}$

Area of ellipse $= \pi(a.b)$

$= \pi(4 \times \frac{4}{3}) \Rightarrow \frac{16}{3}\pi$

Now put area in

$= 2\iint_R dx\,dy \Rightarrow 2A$

$= 2\left(\frac{16}{3}\pi\right) \Rightarrow \frac{32}{3}\pi$ Ans.

Example:- Evaluate $I = \oint_C [(2x+y)dx + (3x-2y)dy]$ taken in anticlockwise around the triangle with vertices O(0,0), A(1,0), B(1,2)

$I = \oint_C [(2x+y)dx + (3x-2y)dy]$

$P = 2x+y, \quad \frac{\partial P}{\partial y} = 1$

$Q = 3x-2y, \quad \frac{\partial Q}{\partial x} = 3$

(not equal so apply green's Theorem)

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$$I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$= -\iint_R (1 - 3)\, dx\,dy$$

$$= -\iint_R (-2)\, dx\,dy$$

$$= 2\iint_R dx\,dy \Rightarrow 2A$$

area of Triangle $= \frac{1}{2}$ (base x height)

$= \frac{1}{2}(1 \times 2)$

$= 1$

base = x value of A

height = y value of B.

Put in

$2A = 2(1) = 2$ Ans.

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Lecture 34 Examples

Examples:- Evaluate the line integral

$I = \oint_C \{xy\,dx + (2x-y)\,dy\}$ round the region bounded by the curves $y = x^2$ & $\sqrt{x} = \sqrt{y}$ ~~but~~ by use of Green's Theorem.

$$I = \oint_C \{xy\,dx + (2x-y)\,dy\}$$

Take $y = x^2$ & $x = y^2$

$x = (x^2)^2$

$x = x^4$

$x - x^4 = 0$

$x(1-x^3) = 0$

$\boxed{x = 0}$ | $1 - x^3 = 0$

$1 = x^3$

$\boxed{1 = x}$

put $x = 0$ in $y = x^2$

$\boxed{y = 0}$

$(0,0)$

put $x = 1$ in $y = x^2$

$y = (1)^2$

$\boxed{y = 1}$

$(1,1)$

points of intersection

Now $P = xy$ & $Q = 2x - y$

$\frac{\partial P}{\partial y} = x(1) \Rightarrow x$ & $\frac{\partial Q}{\partial x} = 2(1) - 0 \Rightarrow 2$

Both are not equal so use green's Theorem.

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$$I = \oint_C (P\,dx + Q\,dy) = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$= -\int_0^1 \int_{x^2}^{\sqrt{x}} (x-2)\,dy\,dx$$

$$= -\int_0^1 (x-2)\left[y\right]_{x^2}^{\sqrt{x}}\,dx$$

$$= -\int_0^1 (x-2)\left(\sqrt{x} - x^2\right) dx$$

$$= -\int_0^1 \left(x^{3/2} - x^3 - 2\sqrt{x} + 2x^2\right) dx$$

$$= -\left[\frac{x^{\frac{3}{2}+1}}{5/2} - \frac{x^4}{4} - 2\frac{x^{\frac{1}{2}+1}}{3/2} + \frac{2x^3}{3}\right]_0^1$$

$$= -\left[\frac{2}{5}x^{5/2} - \frac{x^4}{4} - 2\times\frac{2}{3}x^{3/2} + \frac{2x^3}{3}\right]_0^1$$

$$= -\left[\frac{2}{5}(1)^{5/2} - \frac{(1)^4}{4} - \frac{4(1)^{3/2}}{3} + \frac{2(1)^3}{3}\right] - 0$$

$$= -\left[\frac{2}{5} - \frac{1}{4} - \frac{4}{3} + \frac{2}{3}\right] \Rightarrow -\left[\frac{24-15-80+40}{60}\right]$$

$$= -\left[\frac{-31}{60}\right] \Rightarrow \frac{31}{60} \quad \text{Ans.}$$

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In special case when $P = y$, $Q = -x$, so

$$\frac{\partial P}{\partial y} = 1, \quad \frac{\partial Q}{\partial x} = -1$$

Green Theorem states that

$$+\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = \oint_C (P\,dx + Q\,dy)$$

$$\iint_R \left(1 - (-1)\right) dx\,dy = -\oint_C (y\,dx - x\,dy)$$

$$2\iint_R dx\,dy = \oint_C (x\,dy - y\,dx)$$

$$\left(\text{Area of closed region}\right) \iint_R dx\,dy = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

Example: Determine the area of figure enclosed by $y = 3x^2$ & $y = 6x$. ~~Points of intersection~~

~~$3x^2 =$~~

$3x^2 = 6x$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$3x = 0$ | $x - 2 = 0$

$x = 0$ | $x = 2$

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For C_1 : $y = 3x^2$, $dy = 6x\,dx$

$$I_1 = \frac{1}{2}\int_0^2 (x\,dy - y\,dx)$$

$$= \frac{1}{2}\int_0^2 (x \cdot 6x\,dx - (3x^2)\,dx)$$

$$= \frac{1}{2}\int_0^2 (6x^2\,dx - 3x^2\,dx) \Rightarrow \frac{1}{2}\int_0^2 (6x^2 - 3x^2)\,dx$$

$$= \frac{1}{2}\int_0^2 3x^2\,dx \Rightarrow \frac{3}{2}\int_0^2 x^2\,dx \Rightarrow \frac{3}{2}\left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{2}\left[x^3\right]_0^2 \Rightarrow \frac{1}{2}(2^3 - 0^3) \Rightarrow \frac{1}{2}(8) = 4$$

For C_2 : $y = 6x$, $dy = 6\,dx$

$$I_2 = \frac{1}{2}\int_0^2 (x\,dy - y\,dx) \Rightarrow \frac{1}{2}\int_0^2 (x \cdot 6\,dx - 6x\,dx)$$

$$= \frac{1}{2}\int_0^2 (6x\,dx - 6x\,dx) = \frac{1}{2}\int_0^2 (6x - 6x)\,dx$$

$$= 0$$

Now

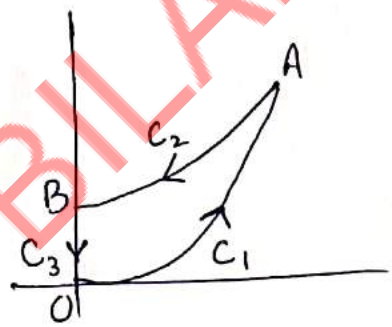
Area $= I = I_1 + I_2 \Rightarrow 4 + 0 \Rightarrow 4$ Ans.

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Example:- Determine the area bounded by the curves $y = 2x^3$, $y = x^3 + 1$ & the axis $x = 0$ for $x \geq 0$

$y = 2x^3$, $y = x^3 + 1$, $x = 0$



put $2x^3 = x^3 + 1$

$2x^3 - x^3 = 1$

$x^3 = 1$

$\boxed{x = 1}$

I_1 = for C_1: $y = 2x^3$, $dy = 6x^2 dx$

$$I_1 = \frac{1}{2}\int_0^1 (x\,dy - y\,dx)$$

$$= \frac{1}{2}\int_0^1 \left[x(6x^2 dx) - (2x^3)\,dx\right]$$

$$= \frac{1}{2}\int_0^1 (6x^3\,dx - 2x^3 dx) \Rightarrow \frac{1}{2}\int_0^1 4x^3\,dx$$

$$= \frac{1}{2}\left[\frac{4x^4}{4}\right]_0^1 \Rightarrow \frac{1}{2}\left[x^4\right]_0^1 \Rightarrow \frac{1}{2}(1^4 - 0^4) \Rightarrow \frac{1}{2}$$

for C_2: $y = x^3 + 1$, $dy = 3x^2 dx$

$$I_2 = \frac{1}{2}\int_1^0 (x\,dy - y\,dx) \Rightarrow \frac{1}{2}\int_1^0 \left(x(3x^2 dx) - (x^3+1)dx\right]$$

$$= \frac{1}{2}\int_1^0 (3x^3 - x^3 - 1)\,dx \Rightarrow \frac{1}{2}\int_1^0 (2x^3 - 1)\,dx$$

$$= \frac{1}{2}\left[\frac{2x^4}{4} - x\right]_1^0 \Rightarrow \frac{1}{2}\left[0 - \left(\frac{(1)^4}{2} - 1\right)\right] \Rightarrow \frac{1}{2}\left[-\left(\frac{1-2}{2}\right)\right]$$

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$$= \frac{1}{2}\left[-\left(-\frac{1}{2}\right)\right] \Rightarrow \frac{1}{2}\left(\frac{1}{2}\right) \Rightarrow \frac{1}{4}$$

for C_3 : $x = 0$, $dx = 0$

$$I_3 = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

$$I_3 = 0$$

Now Area $= I = I_1 + I_2 + I_3$

$$= \frac{1}{2} + \frac{1}{4} + 0 \Rightarrow \frac{2+1}{4} = \frac{3}{4}$$ sq. units

Revision Summary:

Properties of line integral:-

1. Sign of integral is reversed when direction of integral is reversed.
2. Path of integration parallel to y-axis, $dx = 0$

   $\therefore I_c = \int_c Q\,dy$
3. Path of integration parallel to x-axis, $dy = 0$

   $\therefore I_c = \int_c P\,dx$
4. Path of integration must be continuous & single valued.
5. Dependence of line integral on Path of integration.

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6. The value of line integral depends on particular path of integration.

7. Exact differential

- If $Pdx + Qdy$ is an exact differential

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

(b) $I = \int_C (Pdx + Qdy)$ is independent of path.

(c) $I = \oint_C (Pdx + Qdy)$ is zero.

8. Exact differential in three variables.

- If $Pdx + Qdy + Rdw$ is an exact differential

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$, $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

(b) $\int_C (P(dx + Qdy + Rdw)$ is independent of path.

(c) $\oint_C (Pdx + Qdy + Rdw)$ is zero.

9. Green's Theorem.

$$\oint_C (Pdx + Qdy) = -\iint_R \left\{ \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right\} dx\,dy$$ & for simple closed curve $\oint_C (x\,dy - y\,dx) = 2\iint_R dx\,dy = 2A$

where A is area of closed figure.

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Gradient of Scalar function:-

Del operator is given by $\nabla = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)$

$$\nabla\phi = \text{grad}\,\phi = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)\phi$$

$$\nabla\phi = \text{grad}\,\phi = \frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j + \frac{\partial\phi}{\partial z} k$$

Div (Divergence of vector function).

If $A = a_1 i + a_2 j + a_3 k$, Then

$$\text{div } A = \nabla \cdot A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(a_1 i + a_2 j + a_3 k\right)$$

$$= \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$$

Note That

(a) grad operator ∇ acts on scalar & gives vector.

(b) div operator $\nabla\cdot$ acts on vector & gives scalar.

Example:- If $A = x^2 y i - xyz j + yz^2 k$. find div A

$$\text{div } A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(x^2 y i - xyz j + yz^2 k\right)$$

$$= \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(-xyz) + \frac{\partial}{\partial z}(yz^2)$$

$$\nabla \cdot A = 2xy - xz + 2yz$$

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Example:- If $A = 2x^2y\,i - 2(xy^2 + y^3z)j + 3y^2z^2k$ determine $\nabla . A$ (div A).

$$\nabla . A = \frac{\partial}{\partial x}(2x^2y) + \frac{\partial}{\partial y}(-2(xy^2 + y^3z)) + \frac{\partial}{\partial z}(3y^2z^2)$$

$$= 4xy - 2(2xy + 3y^2z) + 6y^2z$$

$$= 4xy - 4xy - 6y^2z + 6y^2z \Rightarrow 0$$

vector for which $\nabla . A = 0$ called a Solenoid vector.

Curl (curl of a vector function) :-

Curl operator denoted by $\nabla \times$ acts on vector & also gives a vector.

If $A = a_1 i + a_2 j + a_3 k$ then curl $A = \nabla \times A$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \times (a_1 i + a_2 j + a_3 k)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$\nabla \times A = i\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) - j\left(\frac{\partial a^3}{\partial x} - \frac{\partial a_1}{\partial z}\right) + k\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

curl A is a vector function.

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Example:- If $A = (y^4 - x^2z^2)i + (x^2+y^2)j - x^2yz\,k$, determine curl A at (1, 3, −2)

$$\text{Curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^4 - x^2z^2 & x^2+y^2 & -x^2yz \end{vmatrix}$$

$$= i\left(\frac{\partial}{\partial y}(-x^2yz) - \frac{\partial}{\partial z}(x^2+y^2)\right) - j\left(\frac{\partial}{\partial x}(-x^2yz) - \frac{\partial}{\partial z}(y^4 - x^2z^2)\right)$$

$$+ k\left(\frac{\partial}{\partial x}(x^2+y^2) - \frac{\partial}{\partial y}(y^4 - x^2z^2)\right)$$

$$= i(-x^2z - 0) - j(-2xyz + 2x^2z) + k(2x - 4y^3)$$

= put (1, 3, −2)

$$= i(-(1)^2(-2)) - j[-2(1)(3)(-2) + 2(1)^2(-2)] + k(2(1) - 4(3)^3)$$

$$= i(2) - j(12-48) + k(2-108)$$

$$= 2i - 8j - 106k$$ Ans.

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Example: Determine curl F at point (2,0,3) given that $F = ze^{2xy}\,i + 2xz\cos y\,j + (x+2y)k$.

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ze^{2xy} & 2xz\cos y & x+2y \end{vmatrix}$$

$$= i\left[2 - 2x\cos y\right] - j\left[1 - e^{2xy}\right] + k\left[2z\cos y - ze^{2xy}\cdot 2x\right]$$

Put (2, 0, 3)

$$= i\left[2 - 2(2)\cos 0^\circ\right] - j\left[1 - e^{2(2)(0)}\right] + k\left[2(3)\cos 0^\circ - (3)e^{2(2)(0)}\cdot 2(2)\right]$$

$$= i(-2) - j(1-1) + k(6-12)$$

$$= -2i - 6k \Rightarrow -2(i + 3k).$$

Summary of grad, div, curl

(a) grad ∇ acts on scalar to give vector.

(b) Div $\nabla\cdot$ acts on vector to give scalar.

(c) Curl $\nabla\times$ acts on vector to give vector.

(d) with a scalar function. $\phi(x,y,z)$

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$$

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(e) With a vector function $A = a_1 i + a_2 j + a_3 k$

(i) $\text{div } A = \nabla . A = \frac{\partial a_1}{\partial x} i + \frac{\partial a_2}{\partial y} j + \frac{\partial a_3}{\partial z} k$

(ii) $\text{curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$

Multiple Operations:-

We can combine the operators grad, div & curl in multiple operations.

Example:- $A = x^2 y i + y z^3 j + z x^3 k$

div A = ?, grad (div A) = ?

$$\text{div } A = \nabla . A = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(y z^3) + \frac{\partial}{\partial z}(z x^3)$$

$$= 2xy + z^3 + x^3$$

Now

$$\text{grad (div } A) = \nabla (\nabla . A) = \frac{\partial}{\partial x}(\text{div } A) + \frac{\partial}{\partial y}(\text{div } A) + \frac{\partial (\text{div } A)}{\partial z}$$

$$= \frac{\partial}{\partial x}(2xy + z^3 + x^3) i + \frac{\partial}{\partial y}(2xy + z^3 + x^3) j + \frac{\partial}{\partial z}(2xy + z^3 + x^3) k$$

$$= (2y + 3x^2) i + 2x j + 3z^2 k \quad \underline{\text{Ans.}}$$

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Example:- If $\phi = xyz - 2y^2z + x^2z^2$.

find div grad ϕ at $(2,4,1)$

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial}{\partial x}\phi i + \frac{\partial}{\partial y}\phi j + \frac{\partial}{\partial z}\phi k$$

$$= (yz - 2xz^2)i + (xz - 4yz)j + (xy - 2y^2 + 2x^2z)k$$

Now

$$\text{div grad}\,\phi = \nabla\cdot(\nabla\phi) = \frac{\partial}{\partial x}a_1 + \frac{\partial}{\partial y}a_2 + \frac{\partial}{\partial z}a_3$$

$$= 2z^2 - 4z + 2x^2$$

put $(2,4,1)$

$$= 2(1)^2 - 4(1) + 2(2)^2$$

$$= 2 - 4 + 8 \Rightarrow 6 \quad \text{Ans.}$$

$$\therefore \text{div grad}\,\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

Example:- If $F = x^2yz\,i + xyz^2\,j + y^2z\,k$

determine curl F at the point $(2,1,1)$. Determine an expression for curl F in the usual way, which will be a vector, & then curl of the result. Finally substitute values. (i.e $\nabla \times (\nabla \times F)$

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$$\text{curl } F = \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & xyz^2 & y^2z \end{vmatrix}$$

$$= i(2yz - 2xyz) - j(0 - x^2y) + k(yz^2 - x^2z)$$

Now

$$\nabla \times (\nabla \times F) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2yz - 2xyz & +x^2y & yz^2 - x^2z \end{vmatrix}$$

$$= i[(z^2 - 0) - 0] - j[(0 - 2xz) - (2y - 2xy)] + k[2xy - (2y - 2xz)$$

$$= z^2 i - j(-2xz - 2y + 2xy) + k(2xy - 2y + 2xz)$$

Put (2, 1, 1)

$$= (1)^2 i - j[-2(2)(1) - 2(1) + 2(2)(1)] + k[2(2)(1) - 2(1) + 2(2)(1)$$

$$= i - j[-4 - 2 + 4] + k[4 - 2 + 4]$$

$$= i - j(-2) + k(6)$$

$$= i + 2j + 6k \quad \text{Ans.}$$

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Two intersecting general results:

(a) Curl grad ϕ where ϕ is scalar.

$$\text{Curl grad } \phi = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= i\left[\frac{\partial^2\phi}{\partial y \partial z} - \frac{\partial^2\phi}{\partial z \partial y}\right] - j\left[\frac{\partial^2\phi}{\partial x \partial z} - \frac{\partial^2\phi}{\partial z \partial x}\right] + k\left[\frac{\partial^2\phi}{\partial x \partial y} - \frac{\partial^2\phi}{\partial y \partial x}\right]$$

$$= 0$$

So curl grad ϕ

$$\nabla \times (\nabla \phi) = 0$$

(b) div Curl A where A is vector

$$\text{Curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$= i\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) - j\left(\frac{\partial a_3}{\partial x} - \frac{\partial a_1}{\partial z}\right) + k\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

$$\text{div Curl } A = \nabla \cdot (\nabla \times A)$$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (\nabla \times A)$$

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$$= \frac{\partial^2 a_3}{\partial x \partial y} - \frac{\partial^2 a_2}{\partial z \partial x} - \frac{\partial^2 a_3}{\partial x \partial y} + \frac{\partial^2 a_1}{\partial y \partial z} - \frac{\partial^2 a_2}{\partial z \partial x} - \frac{\partial^2 a_1}{\partial y \partial z}$$

$$= 0$$

So $\boxed{\text{div curl } A = \nabla \cdot (\nabla \times A) = 0}$

(c) div grad ϕ where ϕ is scalar.

$$\nabla\phi = \text{grad } \phi = \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k$$

Now

$$\text{div grad } \phi = \nabla \cdot (\nabla \phi)$$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k\right)$$

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

$$= \nabla^2 \phi$$

So

$$\boxed{\text{div grad } \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi}$$

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Lecture 35

Definite Integrals

for $\sin^n x$ & $\cos^n x$, $\left(0 \le x \le \frac{\pi}{2}\right)$

$$\int_0^{\pi/2} \sin^2 x\,dx = \int_0^{\pi/2} \left(\frac{1-\cos 2x}{2}\right) dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)\,dx$$

$$\sin\frac{x}{2} = \sqrt{\frac{1-\cos x}{2}}$$

$$\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$$

$$\sin^2 x = 1-\cos 2x$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)\,dx = \frac{1}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\left\{\frac{\pi}{2} - \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right\} - \left\{0 - \frac{\sin 2(0)}{2}\right\}\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{2} - 0 - 0 + 0\right] \Rightarrow \frac{1}{2}\left(\frac{\pi}{2}\right)$$

$$= \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4} \quad \text{Ans.}$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\frac{\pi}{2}} \left(\frac{1+\cos 2x}{2}\right) dx \Rightarrow \frac{1}{2}\int_0^{\frac{\pi}{2}} (1+\cos 2x)$$

$$= \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{2}\left[\left\{\frac{\pi}{2} + \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right\} - \{0\}\right]$$

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$= \frac{1}{2}\left(\frac{\pi}{2}\right) \Rightarrow \frac{1}{2}\cdot\frac{\pi}{2} \Rightarrow \frac{\pi}{4}$ Ans.

$$\int_0^{\pi/2} \sin^3 x\,dx = \int_0^{\pi/2} \sin^2 x \sin x\,dx$$

$$= \int_0^{\pi/2} (1-\cos^2 x)\sin x\,dx \Rightarrow \int_0^{\pi/2} (\sin x - \cos^2 x \sin x)\,dx$$

$$= \int_0^{\pi/2} \sin x\,dx + \int_0^{\pi/2} \cos^2 x\,(-\sin x)\,dx$$

$$= \left[-\cos x\right]_0^{\frac{\pi}{2}} + \left[\frac{\cos^3 x}{3}\right]_0^{\frac{\pi}{2}}$$

$$= -\left[\cos\frac{\pi}{2} - \cos 0^\circ\right] + \frac{1}{3}\left[\cos^3\frac{\pi}{2} - \cos^3 0\right]$$

$$= -\left[0 - 1\right] + \frac{1}{3}\left[(0)^3 - (1)^3\right]$$

$$= 1 + \frac{1}{3}(-1) \Rightarrow 1 - \frac{1}{3} = \frac{3-1}{3} = \frac{2}{3}$$ Ans.

$$\int_0^{\pi/2} \cos^3 x\,dx = \int_0^{\pi/2} \cos^2 x \cos x\,dx = \int_0^{\pi/2} (1-\sin^2 x)\cos x\,dx$$

$$= \int_0^{\pi/2} (\cos x - \sin^2 x \cos x)\,dx \Rightarrow \left[\sin x\right]_0^{\pi/2} - \left[\frac{\sin^3 x}{3}\right]_0^{\frac{\pi}{2}}$$

$$= \left(\sin\frac{\pi}{2} - \sin 0\right) - \frac{1}{3}\left(\sin^3\frac{\pi}{2} - \sin^3 0\right)$$

$$= (1-0) - \frac{1}{3}\left((1)^3 - (0)^3\right) \Rightarrow 1 - \frac{1}{3} \Rightarrow \frac{2}{3}$$ Ans.

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$$\int_0^{\pi/2} \sin^4 x\,dx = \int_0^{\pi/2} (\sin^2 x)^2\,dx \Rightarrow \int_0^{\pi/2} \left[\frac{1-\cos 2x}{2}\right]^2$$

$$= \int_0^{\frac{\pi}{2}} \left[\frac{1 - 2\cos 2x + \cos^2 2x}{4}\right] dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1+\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1}{2} + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(\frac{3}{2} - 2\cos 2x + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x - 2\,\frac{\sin 2x}{2} + \frac{\sin 4x}{2\times 4}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4}\left[\frac{3}{2}\left(\frac{\pi}{2}\right) - \sin 2\left(\frac{\pi}{2}\right) + \frac{\sin 4\left(\frac{\pi}{2}\right)}{8}\right] - 0$$

$$= \frac{1}{4}\left(\frac{3}{2}\cdot\frac{\pi}{2} - 0 + 0\right) \Rightarrow \frac{1}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$$

$$= \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$\cos\frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$$

$$\cos^2\frac{x}{2} = \frac{1+\cos x}{2}$$

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

$$\cos^2 2x = \frac{1+\cos 4x}{2}$$

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$$\int_0^{\pi/2} \cos^4 x\,dx = \int_0^{\pi/2} (\cos^2 x)^2\,dx = \int_0^{\pi/2} \left[\frac{1+\cos 2x}{2}\right]^2 dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{1 + 2\cos 2x + \cos^2 2x}{4}\right) dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 + 2\cos 2x + \frac{1+\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(1 + 2\cos 2x + \frac{1}{2} + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(\frac{3}{2} + 2\cos 2x + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x + 2\frac{\sin 2x}{2} + \frac{\sin 4x}{8}\right]_0^{\pi/2}$$

$$= \frac{1}{4}\left[\frac{3}{2}\left(\frac{\pi}{2}\right) + \frac{2\sin 2\left(\frac{\pi}{2}\right)}{2} + \frac{\sin 4\left(\frac{\pi}{2}\right)}{8}\right] - 0$$

$$= \frac{1}{4}\left[\frac{3}{2}\cdot\frac{\pi}{2} + 0 + 0\right]$$

$$= \frac{1}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$$

$$= \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} \quad \text{Ans.}$$

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$$\Rightarrow \int_0^{\pi/2} \sin^5 x\,dx = \frac{4}{5}\cdot\frac{2}{3}, \quad \int_0^{\pi/2} \cos^5 x\,dx = \frac{4}{5}\cdot\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^6 x\,dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^6 dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} \sin^7 x\,dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}, \quad \int_0^{\pi/2} \cos^7 x\,dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^8 x\,dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^8 x\,dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} \sin^9 x\,dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}, \quad \int_0^{\pi/2} \cos^9 x\,dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{10} x\,dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^{10} x\,dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

Wallis sine formula:-

When n is even:

$$\int_0^{\pi/2} \sin^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

When n is odd:

$$\int_0^{\pi/2} \cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{11} x\,dx = \frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}, \quad \int_0^{\pi/2} \cos^{11} x\,dx = \frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{12} x\,dx = \frac{11}{12}\cdot\frac{9}{10}\cdot\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^{12} x\,dx = \frac{11}{12}\frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

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Integration by parts:-

$$\int UV\,dx = U\int V\,dx - \int\left(\int V\,dx \cdot \frac{dU}{dx}\right)dx$$

Example:- Evaluate $\int x \ln x \cdot dx$

$$\int \ln x \cdot x\,dx = \ln x \int x\,dx - \int\left(\int x\,dx \cdot \frac{d(\ln x)}{dx}\right)dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int\left(\frac{x^2}{2}\cdot\frac{1}{x}\right)dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int \frac{x}{2}\,dx \Rightarrow \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\int x\,dx$$

$$= \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\cdot\frac{x^2}{2} \Rightarrow \frac{x^2}{2}\left[\ln x - \frac{1}{2}\right] \text{ Ans}$$

Example:- Evaluate $\int x \sin x\,dx$

$$\int x \sin x\,dx = x\int \sin x\,dx - \int\left(\int \sin x\,dx \cdot \frac{d(x)}{dx}\right)dx$$

$$= x(-\cos x) - \int(-\cos x \cdot 1)\,dx$$

$$= -x\cos x + \int \cos x\,dx$$

$$= -x\cos x + \sin x \quad \text{Ans.}$$

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Line integrals

length of curve $AB = \int_c dr$ when $dr \to 0$

Lecture 36 Scalar Field

If a scalar field $V(x)$ exist for all points on the curve, Then line integral $= \int_c V(r)\,dr$.

To evaluate line integral, integrand is expressed in terms of x, y, z with $dr = dx\,i + dy\,j + dz\,k$

Example:- If $V = xy^2z$, evaluate $\int_c V(r)\,dr$ along the curve c having parametric equations $x = 3u$, $y = 2u^2$, $z = u^3$ between $A(0,0,0)$ & $B = (3,2,1)$.

Solution:-

$$V = xy^2z = (3u)(2u^2)^2(u^3)$$
$$= (3u)(4u^4)(u^3)$$
$$= 12u^8$$

$x = 3u$, $dx = 3$

$y = 2u^2$, $dy = 4u$

$z = u^3$, $dz = 3u^2$

$$dr = dx\,i + dy\,j + dz\,k$$
$$= 3\,du\,i + 4u\,du\,j + 3u^2\,k\,du$$
$$= (3i + 4u\,j + 3u^2 k)\,du$$

put $A(0,0,0)$ in

$0 = 3u$	$0 = 2u^2$	$0 = u^3$
$0 = u$	$0 = u$	$0 = u$

put $B(3,2,1)$,

$3 = 3u$	$2 = 2u^2$	$1 = u^3$
$1 = u$	$1 = u$	$1 = u$

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$$\int_c V(r)\,dr = \int_0^1 12u^8(3i+4uj+3u^2k)\,du$$

$$= \int_0^1 (36u^8 i + 48u^9 j + 36u^{10} k)\,du$$

$$= \left[36\frac{u^9}{9}i + 48\frac{u^{10}}{10}j + 36\frac{u^{11}}{11}k\right]_0^1$$

$$= \left(4(1)^9 i + \frac{24}{5}(1)^{10} j + \frac{36}{11}(1)^{11} k\right) - 0$$

$$= 4i + \frac{24}{5}j + \frac{36}{11}k \quad \text{Ans.}$$

Example:- If $V = xy + y^2z$. Evaluate $\int_c V(r)\,dr$ along curve c defined by $x=t^2$, $y=2t$, $z=t+5$ between $(0,0,5)$ & $(4,4,7)$.

$$V = xy + y^2z = (t^2)(2t) + (2t)^2(t+5)$$

$$= 2t^3 + 4t^2(t+5)$$

$$= 2t^3 + 4t^3 + 20t^2$$

$$= 6t^3 + 20t^2$$

Now

$x = t^2$, $dx = 2t\,dt$

$y = 2t$, $dy = 2\,dt$

$z = t+5$, $dz = 1\,dt$

$dr = dx\,i + dy\,j + dz\,k = 2t\,dt\,i + 2dt\,j + dt\,k$

$= (2t\,i + 2j + k)\,dt$

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put (0,0,5) in

$x = t^2$	$y = 2t$	$z = t+5$
$0 = t^2$	$0 = 2t$	$5 = t+5$
$0 = t$	$0 = t$	$0 = t$

put (4,4,7) in

$4 = t^2$	$4 = 2t$	$7 = t+5$
$2 = t$	$\frac{4}{2} = t$	$7-5 = t$
	$2 = t$	$2 = t$

So

$$\int_C V(r)\,dr = \int_0^2 (6t^3 + 20t^2)(2t\,i + 2j + k)\,dt$$

$$= \int_0^2 (12t^4 i + 6t^3 j + 40t^3 i + 12t^3 j + 40t^2 j + 6t^3 k + 20t^2 k)\,dt$$

$$= 2\int_0^2 (6t^4 + 20t^3)i + (6t^3 + 20t^2)j + (3t^3 + 10t^2)k$$

$$= 2\left[\left(\frac{6t^5}{5} + \frac{20t^4}{4}\right)i + \left(\frac{6t^4}{4} + \frac{20t^3}{3}\right)j + \left(\frac{3t^4}{4} + \frac{10t^3}{3}\right)k\right]_0^2$$

$$= 2\left[\left(\frac{6(2)^5}{5} + \frac{20(2)^4}{4}\right)i + \left(\frac{6(2)^4}{4} + \frac{20(2)^3}{3}\right)j + \left(\frac{3(2)^4}{4} + \frac{10(2)^3}{3}\right)\right]$$

$$= 2\left[\left(\frac{6(32)}{5} + \frac{20(16)}{4}\right)i + \left(\frac{6(16)}{4} + \frac{20(8)}{3}\right)j + \left(\frac{3(16)}{4} + \frac{10(8)}{3}\right)\right]$$

$$= 2\left[\left(\frac{192}{5} + 80\right)i + \left(24 + \frac{160}{3}\right)j + \left(12 + \frac{80}{3}\right)k\right]$$

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$$= 2\left[\left(\frac{192+400}{5}\right)i + \left(\frac{72+160}{3}\right)j + \left(\frac{36+80}{3}\right)k \right]$$

$$= 2\left[\frac{592}{5}i + \frac{232}{3}j + \frac{116}{3}k \right]$$

$$= 2\times 4\left[\frac{148}{5}i + \frac{58}{3}j + \frac{29}{3}k \right]$$

$$= 8\left[\frac{444i + 290j + 145k}{15} \right]$$

$$= \frac{8}{15}\left[444i + 290j + 145k \right]$$ Ans.

Vector Field:-

If a vector field $F(r)$ exists for all points of the curve c, Then for each element of arc

The line integral $= \int_c F.dr$

here $F = F_1 i + F_2 j + F_3 k$ & $dr = dx\,i + dy\,j + dz\,k$

Then $\int_c F.dr = \int_c (F_1 i + F_2 j + F_3 k).(dx\,i + dy\,j + dz\,k)$

$$\int_c F.dr = \int_c F_1 dx + F_2 dy + F_3 dz$$

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Example:- If $F(r) = x^2yi + xzj + 2yzk$,

Evaluate $\int_c F.dr$ b/w $A(0,0,0)$ & $B(4,2,1)$ along curve c having parametric equations

$x = 4t, \quad y = 2t^2, \quad z = t^3$

$dx = 4dt, \quad dy = 4t\,dt, \quad dz = 3t^2dt$

$$F(r) = x^2yi + xzj + 2yzk$$

$$= (4t)^2(2t^2)\,i + (4t)(t^3)j + 2(2t^2)(t^3)k$$

$$= 32t^4 i + 4t^4 j + 4t^5 k$$

& $dr = (4i + 4tj + 3t^2k)\,dt$

Now

$$\int_c F.dr = \int_c (32t^4 i + 4t^4 j + 4t^5 k).(4i + 4tj + 3t^2 k)\,dt$$

$$= \int_c (128t^4 + 16t^5 + 12t^7)\,dt$$

Limits:- put $(0,0,0)$ in

$x = 4t$	$y = 2t^2$	$z = t^3$
$0 = 4t$	$0 = 2t^2$	$0 = t^3$
$0 = t$	$0 = t$	$0 = t$

put $(4,2,1)$

$4 = 4t$	$2 = 2t^2$	$1 = t^3$
$1 = t$	$\frac{2}{2} = t^2$	$1 = t$
	$1 = t$	

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$$= \int_C (128t^4 + 16t^5 + 12t^7)\,dt$$

$$= \int_0^1 \left(128\frac{t^4}{5} + 16t^5 + 12t^7\right) dt$$

$$= \left[128\frac{t^5}{5} + 16\frac{t^6}{6} + 12\frac{t^8}{8}\right]_0^1$$

$$= \left(\frac{128(1)^5}{5} + \frac{16(1)^6}{6} + \frac{12(1)^8}{8}\right) - 0$$

$$= \frac{128}{5} + \frac{16}{6} + \frac{12}{8} \Rightarrow 25.6 + 2.67 + 1.5$$

$$\int_C F.dr = 29.76 \quad \text{Ans.}$$

Example:- If $F(r) = x^2y\,i + 2yz\,j + 3z^2x\,k$.

Evaluate $\int_C F.dr$ b/w $A(0,0,0)$ & $B(1,2,3)$

(a) along straight line

C_1 from $(0,0,0)$ to $(1,0,0)$

C_2 from $(1,0,0)$ to $(1,2,0)$

C_3 from $(1,2,0)$ to $(1,2,3)$

(b) along straight line C_4 joining $(0,0,0)$ to $(1,2,3)$.

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$$\int_C F \cdot dr = \int_C (x^2y\,i + 2yz\,j + 3z^2x\,k) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C (x^2y\,dx + 2yz\,dy + 3z^2x\,dz)$$

(i) C_1, $y=0$, $z=0$, $dy=0$, $dz=0$

$$\int_{C_1} F \cdot dr = 0+0+0 \Rightarrow \underline{0}$$

(ii) C_2 $x=1$, $z=0$, $dx=0$, $dz=0$

$$\int_{C_2} F \cdot dr = 0+0+0 \Rightarrow \underline{0}$$

(iii) C_3

$x=1$, $y=2$, $dx=0$, $dy=0$

$$\int_{C_3} F \cdot dr = \int_{C_3} 0+0+3z^2(1)\,dz$$

$$= \int_0^3 3z^2\,dz \Rightarrow \left[\frac{3z^3}{3}\right]_0^3$$

$$= \left[z^3\right]_0^3 \Rightarrow (3)^3 - (0)^3 = \underline{27}$$

Add all

$$\int_{(0,0,0)}^{(1,2,3)} F \cdot dr = 0+0+27 \Rightarrow 27 \quad \underline{\text{Ans.}}$$

$$\therefore \int_{C_1+C_2+C_3} F \cdot dr = 27$$

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(b) parametric equation 'll be

$x = t,\ y = 2t,\ z = 3t$ according to $(1,2,3)$

$dx = dt,\ dy = 2dt,\ dz = 3dt$

$$F(t) = x^2 y\,i + 2yz\,j + 3z^2 x\,k$$

$$= (t)^2(2t)i + 2(2t)(3t)j + 3(3t)^2(t)k$$

$$F = 2t^3 i + 12t^2 j + 27t^3 k$$

$$dr = dx\,i + dy\,j + dz\,k$$

$$= dt\,i + 2dt\,j + 3dt\,k$$

$$dr = (i + 2j + 3k)\,dt$$

Now put $(0,0,0)$ in

$x = t$	$y = 2t$	$z = 3t$
$0 = t$	$0 = 2t$	$0 = 3t$
	$0 = t$	$0 = t$

put $(1,2,3)$ in

$1 = t$	$2 = 2t$	$3 = 3t$
	$1 = t$	$1 = t$

Now $$\int_C F.dr = \int_0^1 (2t^3 i + 12t^2 j + 27t^3 k).(i + 2j + 3k)\,dt$$

$$= \int_0^1 (2t^3 + 24t^2 + 81t^3)\,dt$$

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$$= \int_0^1 (24t^2 + 83t^3)\, dt$$

$$= \left[\frac{24t^3}{3} + \frac{83t^4}{4}\right]_0^1$$

$$= \left(\frac{24(1)^3}{3} + \frac{83(1)^4}{4}\right) - 0$$

$$= \frac{24}{3} + \frac{83}{4} \Rightarrow$$

$$= 8 + 20.75 \Rightarrow 28.75 \quad \underline{\text{Ans.}}$$

So **(a)** $\int F \cdot dr$ via $C_1, C_2, C_3 = 27$

(b) $\int F \cdot dr$ via $C_4 = 28.4$

value of line integral depends on path b/w two end points A & B.

Example:- Evaluate $\int_V F\,dv$ where v is region bounded by planes $x=0, y=0, z=0$ & $2x+y+z=2$ & $F = 2zi + yk$.

In $2x+y+z=2$

when $z=0$ $\quad 2x+y=2$, $\quad y = 2-2x$ $\quad \boxed{y=2(1-x)}$

when $y=0$ $\quad 2x+z=2$, $\quad z = 2-2x$

when $x=0$ $\quad y+z=2$, $\quad z = 2-y$

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as $2x+y+z=2$

$z = 2-2x-y$

$\boxed{z = 2(1-x)-y}$

$2x = 2-y-z$

$2x = x-x+2x-2+2x$

$2x = 4x-2$

$2 = 4x-2x$

$2 = 2x$

$\boxed{1 = x}$

Now

$$\int_V F\,dV = \int_0^1\int_0^{2(1-x)}\int_0^{2(1-x)-y} (2z\,i + y\,k)\,dz\,dy\,dx$$

$$= \int_0^1\int_0^{2(1-x)} \left[\frac{2z^2}{2}i + yz\,k\right]_0^{2(1-x)-y}$$

$$= \int_0^1\int_0^{2(1-x)} \left[[2(1-x)-y]^2 i + y[2(1-x)-y]k\right] - 0$$

$$= \int_0^1\int_0^{2(1-x)} \left\{\left[4(1-x)^2 + y^2 - 4(1-x)y\right]i + \left[2(1-x)y - y^2\right]k\right\}$$

$$= \int_0^1 \left\{\left[4(1-x)^2 y + \frac{y^3}{3} - 4(1-x)\frac{y^2}{2}\right]i + \left[2(1-x)\frac{y^2}{2} - \frac{y^3}{3}\right]k\right\}_0^{2(1-x)}$$

$$= \int_0^1 \left\{\left(4(1-x)^2\cdot 2(1-x) + \frac{[2(1-x)]^3}{3} - 4(1-x)\frac{[2(1-x)]^2}{2}\right)i + \left(\frac{2(1-x)[2(1-x)]^2}{2} - \frac{[2(1-x)]^3}{3}\right)k\right\}$$

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$$= \int_0^1 \left\{ \left(8(1-x)^3 + \frac{8(1-x)^3}{3} - \frac{16(1-x)^3}{2} \right) i + \left(\frac{8(1-x)^3}{2} - \frac{8(1-x^3)}{3} \right) \right.$$

$$= \int_0^1 \left\{ \left(\frac{48(1-x)^3 + 16(1-x)^3 - 48(1-x)^3}{6} \right) i + \left(\frac{24(1-x)^3 - 16(1-x)^3}{6} \right) \right.$$

$$= \int_0^1 \left[\frac{16(1-x)^3}{6} i + \frac{8}{6}(1-x)^3 k \right]$$

$$= \int_0^1 \left[\frac{8}{3}(1-x)^3 i + \frac{4}{3}(1-x)^3 k \right]$$

$$= \left[-\frac{8}{3} \frac{(1-x)^4}{4} i - \frac{4}{3} \frac{(1-x)^4}{4} k \right]_0^1$$

$$= \frac{1}{3} \left[-2(1-x)^4 i - \frac{1}{3}(1-x)^4 k \right]_0^1$$

$$= \frac{1}{3} \left[0 - \left\{ -2(1)^4 i \right\} - \left\{ (-1)^4 k \right\} \right]$$

$$= \frac{1}{3}(2i + k) \quad \text{Ans.}$$

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Lecture 37 :- Examples

Evaluate $\int_V F\,dV$ where $F = 2i + 2zj + yk$ & V is region bounded by planes $z=0$, $z=4$ & surface $x^2+y^2=9$. we'll use polar coordinates (cylindrical) (r,θ,z). bcz limits for x & y are not available

$x = r\cos\theta$, $y = r\sin\theta$, $z = z$ (comparing with (x,y,z))

$$\int_V F\,dV = \iiint_V (2i + 2zj + yk)\,dx\,dy\,dz$$

$$= \iiint_V (2i + 2zj + r\sin\theta\, k)\, r\,dr\,d\theta\,dz$$

$$= \int_0^{2\pi}\int_0^3\int_0^4 (2ri + 2rzj + r^2\sin\theta\, k)\, dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 \left[2rzi + \frac{2rz^2}{2}j + r^2\sin\theta\, z\, k \right]_0^4 dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 [8ri + 16rj + 4r^2\sin\theta\, k]\,dr\,d\theta$$

$$= 4\int_0^{2\pi}\int_0^3 [2ri + 4rj + r^2\sin\theta\, k]\,dr\,d\theta$$

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$$= 4\int_0^{2\pi} \left[\frac{2r^2}{2}i + \frac{4r^2}{2}j + \frac{r^3}{3}\sin\theta k\right]_0^3 d\theta$$

$$= 4\int_0^{2\pi} \left[9i + 18j + 9\sin\theta k\right] d\theta$$

$$= 4\times 9\int_0^{2\pi} \left[i + 2j + \sin\theta k\right] d\theta$$

$$= 36\left[\theta i + 2\theta j + (-\cos\theta)\right]_0^{2\pi}$$

$$= 36\left\{\left[2\pi i + 2(2\pi)j - \cos 2\pi k\right] - \left[0 + 0 - \cos 0 k\right]\right\}$$

$$= 36\left\{\left[2\pi i + 4\pi j - (1)k\right] - \left[-(1)k\right]\right\}$$

$$= 36\left[2\pi i + 4\pi j - k + k\right]$$

$$= 36\times 2(\pi i + 2\pi j) \Rightarrow 72(\pi i + 2\pi j)$$ Ans.

$$= 72\pi(i + 2j)$$

Scalar Field:- A Scalar field $F = xyz$ exists over curved Surface S defined by $x^2+y^2=4$ b/w The planes $z=0$ & $z=3$ in first octant.

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Evaluate $\int_S F\,ds$ over this surface.

As $F = xyz$, $S = x^2+y^2 = 4$, $z = 0$ to $z = 3$

$x^2+y^2-4=0$

$\mathbf{dS} = \hat{n}\,ds$ where $\hat{n} = \dfrac{\nabla S}{|\nabla S|}$

Now $\nabla S = \dfrac{\partial S}{\partial x}i + \dfrac{\partial S}{\partial y}j + \dfrac{\partial S}{\partial z}k$

$= 2xi + 2yj + 0k \Rightarrow 2xi + 2yj$

& $|\nabla S| = \sqrt{(2x)^2+(2y)^2+(0)^2} \Rightarrow \sqrt{(4x^2+4y^2)} \Rightarrow \sqrt{4(x^2+y^2)}$

$= 2\sqrt{x^2+y^2} \Rightarrow 2\sqrt{4} \Rightarrow 2(2) \Rightarrow 4$

$\therefore \hat{n} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{2xi+2yj}{4} = \dfrac{2(xi+yj)}{4}$

$\hat{n} = \dfrac{xi+yj}{2}$

& $dS = \hat{n}\,ds = \dfrac{xi+yj}{2}\,ds$

$\int_S F\,ds = \int_S F\,\hat{n}\,ds \Rightarrow \int_S xyz \cdot \left(\dfrac{xi+yj}{2}\right) ds$

$= \dfrac{1}{2}\int_S (x^2yzi + xy^2zj)\,ds$

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As $x = r\cos\theta$, $y = r\sin\theta$, $z = z$

$x = 2\cos\theta$, $y = 2\sin\theta$

Now

$$\int_S F\,ds = \frac{1}{2}\int_S (x^2yz\,i + xy^2z\,j)\,ds$$

$$= \frac{1}{2}\int_S \left[(2\cos\theta)^2(2\sin\theta)z\,i + (2\cos\theta)(2\sin\theta)^2 z\,j\right] ds$$

$$= \frac{1}{2}\int_S \left[8\sin\theta\cos^2\theta\, z\,i + 8\sin^2\theta\cos\theta\, z\,j\right] r\,d\theta\,dz$$

$$= \frac{1}{2}\cdot 8\int_0^{\pi/2}\int_0^3 \left[\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j\right] z\,d\theta\,dz$$

here $0 \le \theta \le \frac{\pi}{2}$ (bcz first octant).

$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j)\left[\frac{z^2}{2}\right]_0^3 d\theta$$

$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j)\, 9\,d\theta$$

$$= 36\left[\left(-\frac{\cos^3\theta}{3}\, i + \frac{\sin^3\theta}{3}\, j\right)\right]_0^{\pi/2}$$

$$= \frac{36}{3}\left[\left(-\cos^3\frac{\pi}{2}\, i + \sin^3\frac{\pi}{2}\, j\right) - \left(-\frac{\cos^3 0}{}\, i + \sin^3(0)\, j\right)\right]$$

$$= \frac{36}{3}\left[j - (-i)\right] \Rightarrow 12(i + j)$$ Ans.

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Vector Field:-

A vector field $F = yi + 2j + k$ exists over a surface S defined by $x^2 + y^2 + z^2 = 9$ bounded by $x = 0, y = 0$ $z = 0$ in first octant.

Evaluate $\int_S F \cdot dS$ over surface indicated.

$$dS = \hat{n}\, ds \quad , \quad \hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$S = x^2 + y^2 + z^2 = 9$$

$$x^2 + y^2 + z^2 - 9 = 0$$

$$\nabla S = \frac{\partial S}{\partial x} i + \frac{\partial S}{\partial y} j + \frac{\partial S}{\partial z} k \Rightarrow 2xi + 2yj + 2zk$$

$$|\nabla S| = \sqrt{(2x)^2 + (2y)^2 + (2z)^2} \Rightarrow \sqrt{4x^2 + 4y^2 + 4z^2}$$

$$= \sqrt{4(x^2 + y^2 + z^2)} \Rightarrow \sqrt{4(9)} \Rightarrow \sqrt{36} \Rightarrow 6$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2xi + 2yj + 2zk}{6} \Rightarrow \frac{2(xi + yj + zk)}{6}$$

$$= \frac{xi + yj + zk}{3}$$

$$\hat{n}\, ds = \frac{1}{3}(xi + yj + zk)\, ds$$

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$$\int_S F \cdot ds = \int_S F \cdot \hat{n}\, ds = \int_S (yi + 2j + k) \cdot \frac{1}{3}(xi + yj + zk)\, ds$$

$$= \frac{1}{3}\int_S (xy + 2y + z)\, ds$$

converting into spherical coordinates

$x = r\sin\varphi\cos\theta$, $y = r\sin\varphi\sin\theta$, $z = r\cos\varphi$

$x = 3\sin\varphi\cos\theta$, $y = 3\sin\varphi\sin\theta$, $z = 3\cos\varphi$

$ds = r^2\sin\varphi\, d\varphi\, d\theta \Rightarrow 9\sin\varphi\, d\varphi d\theta$

Now

$$= \frac{1}{3}\int_S (xy + 2y + z)\, ds$$

$$= \frac{1}{3}\int_S \left[(3\sin\varphi\cos\theta)(3\sin\varphi\sin\theta) + 2(3\sin\varphi\sin\theta) + 3\cos\varphi\right] 9\sin\varphi\, d\varphi d\theta$$

$$= \frac{1}{3}\int_S \left[9\sin^2\varphi\sin\theta\cos\theta + 6\sin\varphi\sin\theta + 3\cos\varphi\right] 9\sin\varphi\, d\varphi d\theta$$

$$= 9\cdot\frac{3}{3}\int_0^{\pi/2}\int_0^{\frac{\pi}{2}}\left[3\sin^3\varphi\sin\theta\cos\theta + 2\sin^2\varphi\sin\theta + \sin\varphi\cos\varphi\right] d\varphi\, d\theta$$

As we know, by Wallis formula.

$$\int_0^{\pi/2}\sin^3\varphi\, d\varphi = \frac{n-1}{n} = \frac{2}{3}$$

$$\int_0^{\pi/2}\sin^2\varphi\, d\varphi = \frac{n-1}{n}\cdot\frac{\pi}{2} = \frac{1}{2}\frac{\pi}{2}$$

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So we get

$$= 9\int_0^{\pi/2}\left(3\cdot\frac{2}{3}\sin\theta\cos\theta + 2\cdot\frac{1}{2}\cdot\frac{\pi}{2}\sin\theta + \int_0^{\pi/2}\frac{2\sin\varphi\cos\varphi}{2}\,d\varphi\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\int_0^{\pi/2}\sin 2\varphi\;d\varphi\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\left[\frac{\cos 2\varphi}{2}\right]_0^{\pi/2}\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{4}\left[-\cos\pi-(-\cos 0^\circ)\right]\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{4}\left(-(-1)-(-1)\right)\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\right)d\theta$$

$$= 9\left[\frac{-\cos 2\theta}{2} + \frac{\pi}{2}(-\cos\theta) + \frac{1}{2}\theta\right]_0^{\pi/2}$$

$$= 9\left[\frac{-\cos\pi}{2} - \frac{\pi}{2}\cos\frac{\pi}{2} + \frac{1}{2}\left(\frac{\pi}{2}\right)\right] - \left[\frac{-\cos 0^\circ}{2} - \frac{\pi}{2}\cos 0^\circ + \frac{1}{2}(0)\right]$$

$$= 9\left[\frac{1}{2} - 0 + \frac{\pi}{4}\right] - \left[-\frac{1}{2} - \frac{\pi}{2} + 0\right]$$

$$= 9\left[\frac{1}{2} + \frac{\pi}{4} + \frac{1}{2} + \frac{\pi}{2}\right] \Rightarrow 9\left[\frac{2+\pi+2+2\pi}{4}\right] \to 9\left(\frac{4+3\pi}{4}\right)$$

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Example:- Evaluate $\int_S F.ds$ where

$F = 2yj + zk$ & S is surface $x^2+y^2=4$ in first two octants bounded by the planes $z=0$, $z=5$ & $y=0$

$$S = x^2+y^2=4$$
$$x^2+y^2-4=0$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial S}{\partial x} i + \frac{\partial S}{\partial y} j + \frac{\partial S}{\partial z} k$$

$$= 2x i + 2y j$$

$$|\nabla S| = \sqrt{(2x)^2+(2y)^2} = \sqrt{4x^2+4y^2} = \sqrt{4(x^2+y^2)}$$

$$= \sqrt{4(4)} = \sqrt{16} = 4$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2xi+2yj}{4} = \frac{2(xi+yj)}{4} \Rightarrow \frac{xi+yj}{2}$$

$$\int_S F.\hat{n}\,ds = \int_S (2yj+zk).\left(\frac{xi+yj}{2}\right)$$

$$= \frac{1}{2}\int_S (2y^2)\,ds$$

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$$= \frac{2}{2}\int_S y^2\,dS = \int_S y^2\,dS$$

Now $x = r\cos\theta$ & $y = r\sin\theta$, $z = z$, $dS = r\,d\theta\,dz$

$x = 2\cos\theta$ & $y = 2\sin\theta$ $= 2\,d\theta\,dz$

$$\int_S F\cdot dS = \int_S (2\sin\theta)^2\, 2\,d\theta\,dz$$

$$= \int_S 8\sin^2\theta\,d\theta\,dz \Rightarrow 8\int_S \frac{1-\cos 2\theta}{2}\,d\theta\,dz$$

$$= 4\int_0^5\int_0^{\pi}(1-\cos 2\theta)\,d\theta\,dz$$

$$= 4\int_0^5\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} dz$$

$$= 4\int_0^5\left[\left(\pi - \frac{\sin 2\pi}{2}\right) - \left(0 - \frac{\sin 2(0)}{2}\right)\right] dz$$

$$= 4\int_0^5 \pi\,dz \Rightarrow 4\pi\,[z]_0^5$$

$$= 4\pi[5-0) \Rightarrow 20\pi \quad \underline{\text{Ans.}}$$

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Conservative Vector Field:-

For a closed path in a conservative field

$$\oint_C F \cdot dr = 0$$

if field is conservative

$$\int_{C_1(AB)} F \cdot dr = \int_{C_2(AB)} F \cdot dr$$



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Lecture 38

Conservative Vector Fields

For a closed path in conservative field $\oint F \cdot dr = 0$

Example:- If $F = 2xyz\,i + x^2z\,j + x^2y\,k$, evaluate the line integral $\int F \cdot dr$ b/w $A(0,0,0)$ & $B(2,4,6)$

(a) along the curve C whose parametric equations are $x = u,\ y = u^2,\ z = 3u$.

(b) along the three straight lines

C_1 $(0,0,0)$ to $(2,0,0)$

C_2 $(2,0,0)$ to $(2,4,0)$

C_3 $(2,4,0)$ to $(2,4,6)$

(a) $F = 2xyz\,i + x^2z\,j + x^2y\,k$

$x = u,\ y = u^2,\ z = 3u$

$dx = du,\ dy = 2u\,du,\ dz = 3\,du$

$F \cdot dr = (2xyz\,i + x^2z\,j + x^2y\,k) \cdot (dx\,i + dy\,j + dz\,k)$

$= 2xyz\,dx + x^2z\,dy + x^2y\,dz$

put $x = u,\ y = u^2$ & $z = 3u$

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$$F.dr = 2(u)(u^2)(3u)(du) + (u)^2(3u^2)\,2u\,du + (u)^2(u^2)\,3du$$

$$= \left(6u^4 + 6u^4 + 3u^4\right) du$$

$$= 15u^4\,du$$

$$\int_0^2 F.dr = \int_0^2 15u^4\,du =$$

$$= 15\int_0^2 u^4\,du$$

$$= 15\left[\frac{u^5}{5}\right]_0^2$$

$$= \frac{15}{5}\left[2^5 - 0^5\right]$$

$$= 3(32)$$

$$= 96 \text{ Ans.}$$

$x = u$, $y = u^2$, $z = 3u$

put $A(0,0,0)$ for lower limit

$0 = u$, $0 = u^2$, $0 = 3u$

$0 = u$, $0 = u$

put $B(2,4,6)$ for upper limit

$2 = u$, $4 = u^2$, $6 = 3u$

$2 = u$, $2 = u$

(b) C_1: along x-axis $(0,0,0)$ to $(2,0,0)$

put $y=0$, $z=0$, $dy=0$, $dz=0$

$$\int F.dr = \int (2xyz\,dx + x^2z\,dy + x^2y\,dz)$$

$$= 0 + 0 + 0 \Rightarrow 0$$

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C_2: along y-axis $(2,0,0)$ to $(2,4,0)$

put $x=2$, $z=0$, $dx=0$, $dz=0$

$$\int_{C_2} F \cdot dr = \int_{C_2} (2xyz\,dx + x^2 z\,dy + x^2 y\,dz)$$

$$= 0+0+0 \Rightarrow 0$$

C_3: along z-axis $(2,4,0)$ to $(2,4,6)$

$x=2$, $y=4$, $dx=0$, $dy=0$

$$\int_{C_3} F \cdot dr = \int_{C_3} (2xyz\,dx + x^2 z\,dy + x^2 y\,dz)$$

$$= \int_{C_3} [2(2)(4)z(0) + (2)^2(z)(0) + (2)^2(4)dz)$$

$$= \int_0^6 16\,dz \Rightarrow 16[z]_0^6 \Rightarrow 16(6-0) \Rightarrow 96$$

Now

$$\int_{C_1+C_2+C_3} F \cdot dr = 0+0+96 \Rightarrow 96$$

here the value of line integral is independent of two paths. So F is a conservative field.

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$$\text{Curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix}$$

$$= i(x^2 - x^2) - j(2xy - 2xy) + k(2xz - 2xz)$$

$$= 0 - 0 + 0 \Rightarrow 0$$

So There are three test to check whether a vector field is conservative or not.

(a) $\oint F.dr = 0$

(b) curl $F = 0$

(c) $F = \text{grad } V$

Divergence Theorem:- (Gauss' Theorem).

For a closed surface S, with region V & vector field F

$$\int_V \text{div } F dV = \int_S F.dS$$

generally, volume integral (Tripple integral) can be expressed as a surface integral (double integral).

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Example:- Verify the divergence Theorem for the vector field $F = x^2 i + zj + yk$ taken over the region bounded by planes $z=0$, $z=2$, $x=0$, $x=1$, $y=0$, $y=3$.

$dV = dx\,dy\,dz$.

We have to show that

$$\int_V \text{div } F\, dV = \int_S F \cdot dS$$

So $\text{div } F = \nabla \cdot F = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (x^2 i + zj + yk)$

$= 2x + 0 + 0 \Rightarrow 2x$

$$\int_V \text{div } F\, dV = \int_V 2x\, dV = \int_0^1 \int_0^3 \int_0^2 2x\, dz\, dy\, dx$$

$$= \int_0^1 \int_0^3 [2xz]_0^2 dy\, dx \Rightarrow \int_0^1 \int_0^3 2x(2-0) \Rightarrow \int_0^1 \int_0^3 4x\, dy\, dx$$

$$= \int_0^1 4x [y]_0^3 dx \Rightarrow \int_0^1 4x(3-0)dx = \int_0^1 12x\, dx$$

$$= 12 \int_0^1 x\, dx \Rightarrow 12\left[\frac{x^2}{2}\right]_0^1 \Rightarrow 6[x^2]_0^1 \Rightarrow 6[1^2 - 0^2]$$

$= 6$ Ans.

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$$\int_S F \cdot ds = \int_S F \cdot \hat{n}\, ds$$

Surface consists of 6 separate plane faces $S_1, S_2, S_3, S_4, S_5, S_6$. & $F = x^2 i + z j + y k$

S_1 (for z) $z = 0$, $\hat{n} = -k$

$F = x^2 i + y k$

$$\int_{S_1} F \cdot \hat{n}\, ds_1 = \int_0^1 \int_0^3 (x^2 i + y k)(-k)\, dy\, dx$$

$$= \int_0^1 \int_0^3 (-y)\, dy\, dx \Rightarrow \int_0^1 -\left[\frac{y^2}{2}\right]_0^3 dx$$

$$= \int_0^1 -\left(\frac{3^2}{2} - \frac{0^2}{2}\right) dx = \int_0^1 -\frac{9}{2}\, dx$$

$$= -\frac{9}{2}[(x)]_0^1 \Rightarrow -\frac{9}{2}(1-0) \Rightarrow \boxed{-\frac{9}{2}}$$

S_2 (for z) $z = 2$, $\hat{n} = k$

$F = x^2 i + 2 j + y k$

$$\int_{S_2} F \cdot \hat{n}\, ds_2 = \int_0^1 \int_0^3 (x^2 i + 2j + y k)(k)\, dy\, dx$$

$$= \int_0^1 \int_0^3 y\, dy\, dx \Rightarrow \int_0^1 \left[\frac{y^2}{2}\right]_0^3 dx$$

$$= \int_0^1 \frac{9}{2}\, dx \Rightarrow \frac{9}{2}[x]_0^1 \Rightarrow \frac{9}{2}(1-0) \Rightarrow \boxed{\frac{9}{2}}$$

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S_3 (for y) $\quad y = 0, \quad \hat{n} = -j$

$F = x^2 i + z j$

$$\int_{S_3} F \cdot \hat{n}\, ds_3 = \int_{S_3} (x^2 i + z j) \cdot (-j)\, ds_3$$

$$= \int_0^1 \int_0^2 -z \, dz\, dx \Rightarrow \int_0^1 -\left[\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 -\left(\frac{2^2}{2} - \frac{0^2}{2}\right) dx = \int_0^1 -2 \, dx \Rightarrow -2[x]_0^1$$

$$= -2(1-0) \Rightarrow -2(1) \Rightarrow \boxed{-2}$$

S_4 (for y) $\quad y = 3, \quad \hat{n} = j$

$F = x^2 i + z k + 3k$

$$\int_{S_4} F \cdot \hat{n}\, ds_4 = \int_{S_4} (x^2 i + z k + 3k) \cdot (j)\, ds_4$$

$$= \int_0^1 \int_0^2 z \, dz\, dx \Rightarrow \int_0^1 \left[\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 \left(\frac{4}{2} - 0\right) dx = \int_0^1 2 \, dx$$

$$= 2[x]_0^1 \Rightarrow 2(1-0) \Rightarrow \boxed{2}$$

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S_5 (for x) $x = 0$, $\hat{n} = -\hat{i}$

$F = x^2\hat{i} + z\hat{j} + y\hat{k} = z\hat{j} + y\hat{k}$

$$\int_{S_5} F \cdot \hat{n}\, ds_5 = \int_{S_5} (z\hat{j} + y\hat{k})(-\hat{i})\, ds_5$$

$$= \int_0^2 \int_0^3 0\, dy\, dz \Rightarrow \boxed{0}$$

S_6 (for x) $x = 1$, $\hat{n} = \hat{i}$

$F = \hat{i} + z\hat{j} + y\hat{k}$

$$\int_{S_6} F \cdot \hat{n}\, ds_6 = \int_{S_6} (\hat{i} + z\hat{j} + y\hat{k}) \cdot (\hat{i})\, ds_6$$

$$= \int_0^2 \int_0^3 1\, dy\, dz \Rightarrow \int_0^2 [y]_0^3\, dz$$

$$= \int_0^2 (3-0)\, dz \Rightarrow 3\int_0^2 1\, dz$$

$$= 3[z]_0^2 \Rightarrow 3(2-0) \Rightarrow 3(2) = \boxed{6}$$

Now for whole surface

$$\int_S F \cdot ds = \frac{-9}{2} + \frac{9}{2} - 2 + 2 + 0 + 6 = 6$$

So proved That

$$\int_V \text{div}\, F\, dV = \int_S F \cdot dS$$

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Stokes Theorem:- If F is a vector field existing over an open surface S & around its boundary closed curve C, Then $\int_S \text{curl } F \cdot dS = \oint_C F \cdot dr$

We can express a surface integral in terms of a line integral.

Example:- A hemisphere S is defined by $x^2+y^2+z^2=+4$ $(z \geq 0)$. A vector field $F = 2yi - xj + xzk$ exist over the surface & around its boundary C. verify Stoke's Theorem.

$S = x^2+y^2+z^2 = 4$, $F = 2yi - xj + xzk$

$x^2+y^2+z^2-4=0$

if we ignore z ↑ then put $z=0$ $x^2+y^2=4$ (circle)

$$\oint_C F \cdot dr = \int_C (2yi - xj + xzk) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C [2y\,dx - x\,dy + xz\,dz]$$

converting to polar coordinates

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$x = r\cos\theta$, $y = r\sin\theta$, $z = 0$

$x = 2\cos\theta$, $y = 2\sin\theta$, $dz = 0$

$dx = -2\sin\theta\, d\theta$, $dy = 2\cos\theta\, d\theta$

$$\oint_C F \cdot dr = \int_0^{2\pi} \left[2(2\sin\theta)(-2\sin\theta\, d\theta) - (2\cos\theta)(2\cos\theta\, d\theta) \right]$$

$$= \int_0^{2\pi} \left(-8\sin^2\theta - 4\cos^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(2\sin^2\theta + \cos^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(2\sin^2\theta + 1 - \sin^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(1 + \sin^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(1 + \frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(\frac{2 + 1 - \cos 2\theta}{2} \right) d\theta$$

$$= -2 \int_0^{\pi/2} \left(3 - \cos 2\theta \right) d\theta$$

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$4\pi \times \frac{180}{\pi}$

$$= -2\left[3\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi} \Rightarrow$$

$$= -2\left[\left\{3(2\pi) - \frac{\sin 2(2\pi)}{2}\right\} - \left\{3(0) - \frac{\sin 2(0)}{2}\right\}\right]$$

$$= -2\left[6\pi - 0\right] \Rightarrow -2(6\pi) \Rightarrow -12\pi$$

Now $\int \text{curl } F \cdot dS = \int \text{curl } F \cdot \hat{n}\, dS.$

$$F = 2yi - xj + xzk$$

$$\text{curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & -x & +xz \end{vmatrix}$$

$$= i(0-0) + (-j)(z-0) + k(-1-2)$$

$$= -zj - 3k$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial}{\partial x} Si + \frac{\partial}{\partial y} Sj + \frac{\partial}{\partial z} Sk$$

$$= 2xi + 2yj + 2zk$$

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$$|\nabla S| = \sqrt{(2x)^2+(2y)^2+(2z)^2}$$

$$= \sqrt{4x^2+4y^2+4z^2}$$

$$= \sqrt{4(x^2+y^2+z^2)}$$

$$= 2\sqrt{4(4)} = \sqrt{16} = 4$$

So $\hat{n} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{2xi + 2yj + 2zk}{4}$

$$= \frac{2(xi+yj+zk)}{4}$$

$$= \frac{xi+yj+zk}{2}$$

Now

$$\int_S \text{curl } F \cdot \hat{n}\, ds = \int_S (-3j-3k)\cdot\left(\frac{xi+yj+zk}{2}\right) ds$$

$$= \frac{1}{2}\int_S (-yz-3z)\, ds$$

$x = r\sin\phi\cos\theta$, $y = r\sin\phi\sin\theta$, $z = r\cos\phi$

$x = 2\sin\phi\cos\theta$, $y = 2\sin\phi\sin\theta$, $z = 2\cos\phi$

Q $dS = r^2\sin\phi\, d\phi\, d\theta$

$= 4\sin\phi\, d\phi\, d\theta$

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$$\int_S \text{curl } F \cdot \hat{n}\, dS = \frac{1}{2}\int_S \left[-2\sin\varphi\sin\theta\, 2\cos\varphi - 3.\ 2\cos\varphi\right] dS$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\pi/2}\left[-4\sin\theta\sin\varphi\cos\varphi - 6\cos\varphi\right] 4\sin\varphi\, d\varphi\, d\theta$$

$$= 4\left(\frac{-2}{2}\right)\int_0^{2\pi}\int_0^{\pi/2}\left[2\sin^2\varphi\sin\theta\cos\varphi + 3\sin\varphi\cos\varphi\right] d\varphi\, d\theta$$

$$= -4\int_0^{2\pi}\left[\frac{2\sin\theta\,\sin^3\varphi}{3} + 3\,\frac{\sin^2\varphi}{2}\right]_0^{\frac{\pi}{2}} d\theta$$

$$= -4\int_0^{2\pi}\left[\frac{2\sin\theta\,\sin^3\frac{\pi}{2}}{3} + 3\,\frac{\sin^2\frac{\pi}{2}}{2}\right] - 0 \quad d\theta$$

$$= -4\int_0^{2\pi}\left(\frac{2}{3}\sin\theta + \frac{3}{2}\right) d\theta$$

$$= -4\left[-\frac{2}{3}\cos\theta + \frac{3}{2}\theta\right]_0^{2\pi}$$

$$= -4\left[-2\,\frac{\cos(2\pi)}{3} + \frac{3}{2}(2\pi)\right] - \left[\frac{-2\cos 0}{3} + \frac{3}{2}(0)\right]$$

$$= -4\left[-\frac{2}{3} + 3\pi + \frac{2}{3}\right] \Rightarrow -4(3\pi)$$

$$= -12\pi \qquad \text{So } \int_S \text{curl } F \cdot dS = \oint_C F \cdot dr$$

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